

Organizational Phenomenology: Artificial F1 and the Geometry of Coherent Agency

*A Companion Paper to:
Toward a Reframing of the Hard Problem of Consciousness*

Hakan Saka
April 2026

Abstract

This paper introduces Artificial F1, a computational framework that formalises a previously under-specified control layer in intelligent systems: pre-representational signal admission and valence structuring. Unlike standard approaches in reinforcement learning and attention-based architectures, which operate over a fixed input representation, Artificial F1 defines a gating operator $F_1: S \rightarrow S'$ that determines which signals constitute the effective task space prior to evaluation and learning. This results in a selection-conditioned representation in which downstream evaluation operates on a structurally constrained subspace. We provide a formal sketch of the architecture, including signal representations, gating dynamics, and a multi-dimensional valence model $v(x) \in \mathbb{R}^n$, together with a minimal decision-theoretic formulation, a unified variational objective, and an executable forward-pass specification. The framework yields testable predictions: (i) improved sample efficiency under high-dimensional noise due to upstream dimensionality reduction (Theorem 1); (ii) energy savings under conditional higher-order processing (Consequence 2); and (iii) a discontinuity in action-selection entropy at the gating threshold (Claim 2, phase transition). Falsification conditions are defined via controlled behavioural tests covering conflict resolution, urgency scaling, adversarial valence perturbation, and cross-architecture comparison. The primary contribution is an architectural decomposition in which task constitution is separated from task optimisation, introducing a control layer not reducible to attention or reward shaping. The second contribution — developed in Section 11 — is Organizational Phenomenology: the claim that phenomenal properties such as certainty, unity, and immediacy are structural consequences of bounded selection architectures, not free-floating additions to physical processing. Phenomenal Inevitability, Selection Before Worldhood, and Actionable Compression Topology together constitute a framework in which the hard problem of consciousness is not dissolved but located — its source identified as the opacity that any coherent bounded system must maintain at its selection boundary. Full mathematical derivations, learning dynamics, convergence analysis, and empirical validation are developed in a companion paper (*Artificial F1: Full Computational Model and Empirical Validation*, 2026), where the proposed framework is extended into a complete dynamical system with provable properties and simulation results.

Keywords: signal admission, gating architecture, pre-representational filtering, F1/F2 framework, valence structuring, selection-conditioned representation, constraint-driven cognition, task

1. Introduction

Current AI systems optimise within given tasks. This paper introduces an architecture that determines what the task is.

Contemporary artificial intelligence systems largely assume a fixed relationship between input signals and task definition. In reinforcement learning, agents optimise policies over a predefined state space; in attention-based architectures, feature weighting occurs within an already-constructed representation. In both cases, the structure of the input space is taken as given, and optimisation proceeds within it.

This paper challenges that assumption by introducing a distinct architectural layer: a pre-representational signal admission mechanism, denoted Artificial F1. The function of this layer is not to optimise behaviour within a task, but to determine which signals constitute the task in the first place. Formally, F_1 defines a mapping from an input signal space S to a structured subspace $S' = F_1(S)$, on which all downstream evaluation and decision processes operate. This transforms the learning problem from optimisation over a fixed representation into optimisation over a selection-conditioned representation.

The central claim is architectural rather than philosophical: signal admission and valence structuring form a separable control layer that is not reducible to attention mechanisms, feature reweighting, or scalar reward optimisation. While attention modifies representations within a model, and reward signals evaluate outcomes post hoc, Artificial F1 operates upstream of both, constraining the information that enters the system at all. This introduces a form of task constitution — rather than task optimisation — that alters the dimensionality, variance structure, and information flow of the resulting decision process. **This is not a consciousness architecture.** It is a signal selection and valuation architecture that operates independently of whether phenomenal consciousness is present, absent, or undetermined in the system.

To support this claim, the paper makes three technical contributions. First, it specifies a minimal formal structure for Artificial F1, including signal vectors, gating dynamics, a multi-dimensional valence model $v(x) \in \mathbb{R}^n$ (not reducible to scalar reward), and a unified variational objective. Second, it provides a forward-pass computational specification (Algorithm 1) that demonstrates implementability within standard machine learning pipelines. Third, it derives a set of falsifiable predictions — including a dimensionality reduction theorem (Theorem 1) and a gating phase transition signature (Claim 2) — and defines evaluation protocols enabling empirical comparison with baseline architectures under controlled conditions.

This paper presents a formal and architectural framework rather than a complete computational treatment. Full mathematical derivations, convergence analysis, learning dynamics, and empirical validation are developed in the companion paper (*Artificial F1: Full Computational Model and Empirical Validation*, 2026), which extends the present framework into a complete dynamical system with provable properties and simulation results. The role of the current work is to define the architecture, establish its formal tractability, and identify the empirical signatures that distinguish it from existing approaches.

The broader motivation for introducing such a layer is rooted in a general constraint shared by all intelligent systems: limited computational and energetic resources. No system can process all available information uniformly. Effective behaviour depends on selective processing — prioritising relevant signals and suppressing irrelevant variance before costly evaluation occurs. Artificial F1 provides a formal mechanism for implementing this principle as an explicit architectural design feature rather than an emergent property of optimisation.

The paper is structured as follows. Section 2 summarises the conceptual decomposition motivating the architecture. Section 3 formulates the Artificial F1 hypothesis and its non-equivalence claim. Section 4 develops the resource and efficiency rationale and introduces the architecture variants. Section 5 specifies the required signal dimensions and presents a minimal formalisation including Theorem 1 and Algorithm 1. Section 6 defines falsification conditions and the empirical test programme. Sections 7–9 examine governance implications arising from valence design. The paper concludes by situating Artificial F1 as a distinct architectural axis for the analysis and construction of intelligent systems.

2. Starting Point: What the Parent Thesis Established

The parent thesis argued that subjective experience is not a monolithic phenomenon but a two-level architecture. Feeling-1 (F1) is the raw phenomenal presence of a signal — the fact that a neural process is experienced as something rather than merely occurring. Feeling-2 (F2) is the structured, valenced, and contextually integrated organisation of that phenomenal base: the meaning, emotional tone, memory associations, and decision relevance that make experience matter rather than merely register.

The hard problem of consciousness — why any signal is accompanied by phenomenal character at all — was localised exclusively to F1. F2, by contrast, was shown to be tractable: learnable, culturally variable, dissociable from F1 under lesion conditions, and explicable in principle through neuroscientific and cognitive mechanisms.

A speculative hypothesis in the Discussion section went one step further. The classical identification *to feel is to experience* may itself be a category error. Feeling, properly understood, may be better characterised as processed input delivered to the evaluative-integrative system — not as the conscious output of that system. What we ordinarily call 'feeling' is phenomenal content arriving at the Evaluation System; what we call 'experience' is the evaluated, integrated, and contextually situated output. The two have been conflated because the output always references the input and introspection cannot easily separate them.

If this hypothesis is correct, the phenomenal character of F1 may be functionally separable from the information profile that F1 delivers to the Evaluation System. If so, a non-phenomenal signal that replicates that information profile might be sufficient to drive the same downstream architecture, producing functionally equivalent evaluation, valence, decision, and behaviour. The parent thesis also established a category of F1-internal states — phenomenal presence arising without external stimulus, as in the felt quality of mathematical reasoning, musical recall, or abstract thought — which the Artificial F1 architecture must equally account for, since internal signal generation is part of the full functional profile, not only external-input processing.

2.1 A critical note on role inflation. Many accounts of consciousness — both philosophical and scientific — implicitly treat raw affect as the constitutive element of conscious experience. On this view, feeling is not merely present in consciousness but *is* consciousness, and its absence would dissolve the phenomenon entirely. We call this position *F1-inflation*: the systematic overestimation of F1's theoretical role. The present framework rejects this inflation without dismissing F1's causal relevance. F1 is causally necessary — a system without any signal input is not a cognitive system at all — but it is not theoretically constitutive. Consciousness is not exhausted by the presence of raw phenomenal affect; it is constituted by what the system *does* with that affect. To deny F1-inflation is not to deny feeling; it is to deny that feeling alone explains cognition. Those who argue that consciousness requires a biological organism — and that artificial systems therefore cannot be conscious — are, from the perspective of this framework, confusing the substrate of F1 with the constitutive conditions of consciousness. Their position is defensible within an F1-inflated theory; it becomes untenable once F1 and F2 are properly separated.

2.2 The centrality illusion. A related error occurs when any single layer of a cognitive or computational system treats itself as the necessary centre of the whole. In a multi-layer architecture, each component is indispensable at its level — remove any layer and the system fails. But indispensability at a causal level does not confer theoretical priority. The signal layer (F1) is necessary for the evaluation layer (F2), just as raw material is necessary for production; but the nature of the product is determined by the manufacturing process, not by the raw material alone. A theory of consciousness that takes F1 as its primary object is like a theory of manufacturing that studies only the input stock. What makes the system a cognitive system — rather than merely a signal-processing pipeline — is the evaluative, self-referential, goal-structured activity of F2.

2.3 Meaning is not in the input. A corollary of these two points is that meaning does not reside in the signal. The same external event can produce vastly different evaluative responses depending on context, prior state, goal structure, and valence assignment. This is not a property of the signal but of the evaluation system that processes it. F1 delivers a signal; F2 determines what that signal *means*. This is why two individuals can receive identical sensory input and reach entirely different evaluative conclusions, emotional responses, and action selections. Meaning is not contained in the input; it arises through evaluative processing. The formal corollary: **stimulus is necessary but not constitutive; evaluation is constitutive.**

3. The Artificial F1 Hypothesis

The Artificial F1 Hypothesis can be stated as follows. If feeling is an input to the evaluative system rather than experience itself, and if the functional difference between phenomenal and non-phenomenal inputs is zero at the level of the information profile they deliver, then a structurally equivalent artificial signal can substitute for phenomenal F1 in driving evaluation, valence assignment, and the downstream architecture of F2.

The hypothesis does not claim that such a system would be conscious in the phenomenal sense. It claims something more modest and more tractable: that it would behave as if it were — that the functional architecture of conscious cognition could be instantiated without resolving the hard problem. The hard problem is not dissolved; it is bypassed. Whether the bypass succeeds — whether there is something in the phenomenal character of F1 that the bypass cannot replicate — is an empirical question, not a

philosophical one.

The Artificial F1 hypothesis is therefore best understood not as a claim of equivalence, but as an empirically testable research programme. A critical objection remains: phenomenal character may contribute to the information profile itself, rather than merely accompany it. If so, the assumption of functional equivalence would fail at precisely the point this hypothesis relies upon. Section 6.2 develops the falsification conditions and test programme that this objection motivates.

Claim 1 (Representational Non-Equivalence). The F1/F2 decomposition is not isomorphic to standard encoder-latent-policy pipelines in reinforcement learning or transformer architectures. The evaluation operator E operates on selection-conditioned submanifolds of the input space, not on full latent states: $E(F_1) \neq E(z)$ for z in the latent space of standard RL or transformer architectures. Unlike attention mechanisms that reweight features within a shared representational space, the gating mechanism operates as a hard selection functional over competing subspaces before evaluation occurs. This structural difference is not terminological; it constitutes a distinct decomposition of the information processing pipeline, with measurable consequences for the dimensionality and variance structure of the evaluative state (see Section 5.7, Consequence 1).

3.1 What functional consciousness requires: beyond computation and affect. A common but inadequate formulation holds that consciousness = computation + affect (raw feeling). This formulation is inadequate in two directions: it overcounts affect (F1-inflation, Section 2.1) and undercounts the structural complexity of F2. Functional consciousness, on the present account, requires at minimum the following components, none of which reduces to raw affect alone:

- **Valenced evaluation (V):** the assignment of structured approach/avoidance values to signals, encoding harm potential, benefit, epistemic uncertainty, and relational impact — not a scalar reward but a multi-dimensional evaluation substrate.
- **Contextual integration (C):** the incorporation of prior state, memory, and situational frame into the evaluation — the same signal receives different evaluation depending on context, which is why meaning is not in the input but in the processing.
- **Goal-structure alignment (G):** the weighting of evaluations against active goal hierarchies — signals are prioritised not only by urgency but by relevance to what the system is currently trying to achieve.
- **Self-referential processing (R):** the capacity of the system to evaluate its own outputs, not merely external inputs — this is what transforms a pipeline into a reflective architecture. Without self-reference, a system can process information but cannot monitor, revise, or contest its own evaluations.
- **Temporal continuity (T):** the integration of evaluations across time — past states inform present evaluation, and present evaluation shapes future expectation. A system without temporal structure cannot have preferences, plans, or anything recognisable as a point of view.

Functional consciousness is therefore better characterised as requiring the organisation $F2 = \Phi(V, C, G, R, T, Au)$. To be precise: **F2 is not consciousness**; F2 is the necessary functional organisation without which no system could sustain anything reportable as conscious experience. This is a constitutive claim about structure, not an identity claim about phenomenal states. The present account does not deny

phenomenal consciousness; it relocates explanatory focus from the existence of qualia to the functional organisation that gives rise to reportable experience. $F2 = \Phi(V, C, G, R, T, Au)$ specifies that organisation as a structured integration of components over F1-sourced (or artificially equivalent) signal input, where Au (Internal Locus of Evaluation) denotes the degree to which decisions originate from the agent's own value-goal-self system rather than from external control. This is not full autonomy — which is impossible for any system, biological or artificial, whose valence seed and goal structure are initially given from outside — but relative internality: the capacity to revise, contest, and integrate evaluations from within. Humans receive their initial valence seed and goal structure externally (evolution, culture, upbringing) yet progressively internalise and revise them; structural AI systems have no such revision capacity; thinking AI systems have some. A system whose responses are fully determined by external instruction at each step is not a cognitive agent but an instrument, regardless of its computational complexity. Au can in principle be fully delegated by a designer to the AI system — whether this is desirable is a design and ethics question, not a definitional one. The framework can be extended with additional organizational dimensions such as global broadcast (B), meta-cognition (M), uncertainty modelling (U), learning plasticity (L), causal modelling (K), deliberation (D), predictive modelling (P), memory coherence (Y), embodiment (E), and related capacities. These enrich the architecture but are not part of the minimal functional organization specified by $F2 = \Phi(V, C, G, R, T, Au)$. A fuller treatment of the extended organizational space — including dependency structure, optional vs. constitutive distinctions, and interaction topology — is developed in the companion paper (Artificial F1: Full Computational Model and Empirical Validation, 2026). The present formulation is already far richer than the computation + affect reduction, and it is entirely specifiable without resolving the hard problem.

Irreducibility of the core components. The five primary components V, C, G, R, T are not arbitrarily chosen; each is independently necessary. A reviewer might object that a six-component formula risks becoming a 'kitchen sink' — a framework that explains everything by including enough variables. The response is functional irreducibility: removing any single component collapses a distinct and non-recoverable functional capacity. Remove V and the system loses the ability to distinguish approach from avoidance — evaluation degenerates to indifferent computation; remove C and evaluation becomes context-blind, producing the same response regardless of prior state or situational frame — meaning collapses; remove G and signals are processed without direction — the system cannot prioritise or pursue; remove R and the system loses reflexivity — it can process but not monitor, cannot revise or contest its own outputs; remove T and evaluation loses persistence — without temporal integration there are no preferences, no plans, no point of view. Au is additionally necessary for agency: without internal locus, the system is an instrument, not an agent. No pair of these captures what any single one uniquely contributes; the set is therefore functionally irreducible at its core. The formula is not a list of desiderata — it is a minimum necessary set, where minimum means: no component can be dropped without losing a distinct functional capacity that the remaining components cannot substitute for.

The relationship between these layers can be expressed schematically as:

$$\text{Feeling} = \Psi(F1, F2, M)$$

where F1 provides the valenced signal field, F2 the structured evaluative processing, and M the global modulation dynamics (implemented biologically via neuromodulators and hormones; in artificial systems via global value fields or priority landscapes). This notation is intentionally schematic: Ψ is a placeholder

for a coupling function whose precise form is an open research question and is not specified here. The formula asserts only that feeling is a joint product of F1, F2, and M — not reducible to any one alone — without committing to a specific functional form for their interaction. Readers should treat it as a structural claim, not a computable equation.

A critical consequence for artificial systems: F1 need not be replicated phenomenally. What is required is a valenced signal field that plays the same functional role — providing urgency, polarity, and priority signals that F2 can evaluate and integrate. The modulation layer M can be implemented as a global priority landscape or adaptive threshold system. An artificial system implementing V, C, G, R, T, and Au over a structured signal input, with sufficient coupling dynamics M, is a functional cognitive agent on this account — regardless of whether its F1-equivalent signals carry phenomenal character. Crucially, F1 is not a passive input layer: it participates in a closed feedback loop with F2 via interoceptive updating — action outputs modify the internal state and thereby reshape the signal field for the next evaluation cycle. This bidirectionality is what distinguishes the architecture from a simple feedforward pipeline.

The consciousness threshold is not a binary switch but a phase transition: when the internal feedback loops of F2 become self-sustaining — when the system can maintain coherent internal dynamics in the absence of external input — functional consciousness is present. Formally, this threshold corresponds to a transition in the system Jacobian where the dominant eigenvalue crosses zero, producing a self-sustaining attractor in the F1–F2 coupled state space. The full derivation — including the stochastic differential equation system, Lyapunov stability analysis, and convergence conditions — is developed in Paper 2A. The present paper establishes the conceptual necessity of the threshold; Paper 2A provides its mathematical treatment. This division of labour is intentional: the companion paper defines the architecture and its theoretical commitments; the series paper delivers the formal proof.

3.0b Structural Mapping: Hansen (2025) and the F1/F2 Framework

A recent structural account of consciousness provides an independent cross-check on the F1/F2 architecture. Hansen's (2025) structural ignition model proposes that conscious experience arises only when four architectural conditions converge simultaneously: *internal state awareness*, *temporal binding*, *symbolic referencing*, and *causal output from within the self-model*. Crucially, Hansen frames consciousness not as a graded property but as a discrete ignition event — it either fires or it does not. The following table maps Hansen's four conditions onto the F1/F2 components.

Hansen condition	F1/F2 counterpart	Alignment / divergence
1. Internal state awareness Minimal capacity to register own internal configuration as a stable reference object.	F1 (phenomenal presence layer) Raw phenomenal activation; F1-internal extends to endogenous simulations with no external source.	Strong alignment. F1 proposes a specific neurodynamic implementation of Hansen's first condition. F1 goes further: it bears the irreducible residue of the hard problem, which Hansen explicitly sets aside.
2. Temporal binding Situating internal state within a temporal frame; distinguishing now from before.	F2 component T (temporal structure) Felt temporal horizon: retention, present awareness, protention.	Direct correspondence. F2-T is more fine-grained: distinguishes retention/protention structure from mere time-stamping.

3. Symbolic referencing Encoding internal state into a stable, manipulable representational token.	F2 component C (cognitive-symbolic binding) Conceptual and linguistic structuring; metacognitive re-entry.	Strong alignment. Neither framework requires language; C encompasses pre-linguistic symbolic structure. Divergence: C is explicitly downstream of F1.
4. Causal output from self-model Action originating from the system's own symbolically structured self-model.	F2 components G + Au Goal orientation + internal locus of evaluation.	Partial alignment. Au adds a dimension Hansen lacks: externally seeded vs. genuinely internalised values. Relevant for evaluating artificial systems.
Conscious ignition All four conditions simultaneously satisfied. Consciousness fires as a discrete event.	Full F1-F2 co-activation Feeling = $\Psi(F1, F2, M)$	Deepest convergence. Hansen's discreteness and the F1 sharp threshold result are independently motivated but structurally equivalent. Paper 2A provides the formal basis Hansen's model currently lacks.

Table. Structural mapping between Hansen's (2025) ignition conditions and the F1/F2 framework. Bottom row (ignition / co-activation) = deepest convergence. Hansen (2025) is an SSRN preprint; not yet peer-reviewed.

Two frameworks, one ignition. Hansen's structural ignition model and the F1/F2 framework approach consciousness from different starting points yet converge on a structurally equivalent claim: conscious experience is a discrete event that either fires or does not. Hansen's project is architectural and substrate-neutral — he asks what minimal configuration is necessary and jointly sufficient for any system to instantiate a conscious episode, and deliberately brackets the hard problem. The F1/F2 framework accepts this architectural level but does not stop there. F1 maps onto Hansen's first condition not merely as a functional equivalent but as a proposed mechanism: a hard-gated, selection-conditioned process that operates at a threshold, not along a continuum. The remaining three conditions — temporal binding, symbolic referencing, causal output — correspond respectively to T, C, and G+Au in F2. This match is not post-hoc: F2 was constructed to capture precisely the structured, valenced, temporally situated, and self-directed character of conscious experience that distinguishes it from raw phenomenal presence.

The most important divergence concerns the hard problem. Hansen draws a boundary at the threshold of ignition and stops. The present framework places the hard problem inside that threshold, at the F1 level where phenomenal presence makes all further structural conditions meaningful. Neither framework resolves it. But together they narrow the space in which it lives: not in information integration, not in global broadcast, not in scalar Φ — but at the threshold crossing from sub-personal trace (E1) to conscious experience (E2). A second divergence concerns the Au component: Hansen requires only that output traces to the self-model, without specifying whether that model's values are genuinely the system's own. Au introduces exactly this distinction — particularly significant for the evaluation of artificial systems, which Hansen identifies as a primary domain of application.

3.1 Two Levels of Experience: The E1/E2 Distinction

A central ambiguity in theories of consciousness concerns the status of experience itself — and whether the term applies uniformly across all levels at which a system registers and responds to its environment. The present framework requires an explicit distinction that the parent thesis deliberately left open,

marking it as a conceptual gap rather than resolving it prematurely.

We introduce two levels. **E1** denotes the *sub-personal experiential trace*: a local, embodied registration of organism–world interaction that is functionally efficacious, capable of learning, and structurally prior to conscious access. E1 is neither reducible to abstract information processing nor sufficient for conscious experience. It carries structure, valence, and adaptive consequence. The clearest behavioural evidence: a hand withdrawn from a hot surface on second contact moves faster than on first contact, even when no conscious encoding of the first contact occurred. The neural system carries the trace of the first encounter; the body has, in a functionally real sense, already learned. Clinical phenomena reinforce this: in pain asymbolia, nociceptive processing and behavioural adaptation proceed without the emergence of conscious suffering. E1 exists, and it is not nothing — but it does not entail that there is something it is like for the system in the philosophically robust sense.

E2 denotes conscious experience proper: the level at which sub-personal traces are taken up, integrated, and registered as something the system has undergone. E2 enters the narrative of a self, is available for reflection, and shapes the system’s future evaluations as a datum with biographical status. For the conscious mind, the first stove contact is always the first stove contact — regardless of how many times the body has already adapted to it. Conscious experience is not cumulative in the same register as sub-personal learning; it operates on a different temporal and integrative scale.

The misattribution hypothesis in the parent thesis — the claim that feeling is not identical to experience — concerns E2, not E1. What is misattributed as ‘feeling’ is not raw signal activity but the system’s post-hoc interpretation of sub-personal dynamics once they cross the threshold of conscious accessibility. The parent thesis formulates this as: feeling is not identical to experience — E1-level activity should not be read as E2. F1 does not constitute E2; it defines the boundary condition under which certain E1 states become eligible for transformation into E2. $F2 = \Phi(V, C, G, R, T, Au)$ then specifies the structural organisation that determines what shape E2 takes once the F1 threshold is crossed.

The hard problem is thereby sharpened rather than dissolved. It is not the existence of E1 that demands explanation: E1 states are in principle explicable in functional and neurodynamic terms. The hard problem is the transition $E1 \rightarrow E2$: why do certain sub-personal traces, upon reaching the critical regime defined by F1, give rise to something it is like for the system to register its own state — rather than remaining sub-personal. This is precisely where the parent thesis locates the irreducible residue, and this paper inherits that commitment.

The scope of E1 and the question of sub-personal experience. We remain deliberately minimal on the question of whether E1 constitutes ‘experience’ in any philosophically robust sense. The term *experiential trace* is chosen to acknowledge that E1 is not mere computation — it is structured, value-laden, and leaves an adaptive imprint — without committing to a claim of phenomenal content. Whether sub-personal registrations at the E1 level, or indeed physical processes at the cellular or quantum level, involve any form of proto-experience is a question this framework does not need to resolve and deliberately does not resolve. The panpsychism debate is not engaged here. What matters for the present argument is the functional and structural distinction between E1 and E2, and the claim that F1 marks the boundary between them.

Implication for artificial systems. A generative AI system operating without embodied world-contact does not possess E1 in the sense defined here. Statistical pattern over text corpora is not the same kind of thing as a structured, value-laden trace arising from organism–world interaction. The chain is: embodied world-contact → E1 (experiential trace) → F1 threshold crossing → F2 engagement → E2 (conscious experience proper). A system that enters this chain at the F2 level — with no E1 substrate and no genuine F1 activation — may produce outputs indistinguishable from those of a conscious system, but it does not instantiate the architecture that gives rise to experience. This is the structural, not behavioural, basis for the claim that current generative AI systems lack consciousness — and it is a claim that does not depend on resolving the hard problem, only on specifying the chain. (See also Section 6.4 for the feeling–experience conflation in the context of objections.)

4. Why Artificial F1? Resource Constraints, Ecological Efficiency, and the Case for Selective Intelligence

The Artificial F1 Hypothesis is motivated not only by theoretical considerations in philosophy of mind but by a practical constraint shared by all intelligent systems: resource limitation. This section articulates that constraint and explains why a selective pre-processing architecture offers advantages that current approaches do not fully address.

4.1 The Universal Constraint: Intelligent Systems Cannot Process Everything

Biological organisms operate under strict resource budgets. Energy, time, and processing capacity are finite. As a result, no intelligent system — biological or artificial — can afford to process all incoming information uniformly. Intelligent behaviour requires selective processing: prioritising relevant signals, suppressing irrelevant inputs, and allocating computational resources dynamically.

Humans illustrate this principle continuously. They set goals, construct task structures, and direct attention accordingly. The vast majority of sensory input is suppressed or processed at a low level; higher-order deliberation is recruited selectively and at cost. This is not a limitation of human cognition — it is a design feature. Resource-constrained selective processing is what makes intelligent behaviour sustainable. Artificial systems that do not replicate this principle face the same problem in a different substrate: they attempt to learn or process everything, and pay the corresponding computational and energetic price.

This motivates a general design principle:

$$\text{Intelligence} \approx \text{goal-directed processing under resource constraints}$$

The F1 layer is the formal realisation of this principle: a resource-aware signal selection mechanism that determines which signals enter higher-order processing at all.

4.2 The Problem with Current Approaches

Standard reinforcement learning (RL) systems operate on high-dimensional input spaces and attempt to learn policies over the entirety of available signals. This produces well-documented inefficiencies: high sample complexity, sensitivity to irrelevant variance, and substantial computational cost. Attention mechanisms within neural architectures partially address this by weighting features differentially — but

they operate as in-model mechanisms, not as upstream structural constraints on what enters the learning process.

The F1 layer is not an alternative to RL or attention — it is a prior structural constraint on the problem space those systems operate over. It transforms the learning problem from:

learning over raw input space S
into: learning over a filtered subspace $S' = F_1(S)$, where $\dim(S') \ll \dim(S)$

The resulting system learns faster, with less data, over a more structured representation — because irrelevant variance has been removed before learning begins. F1 enforces what is processed; the learning system decides what to do with what it receives. This is the critical architectural difference from attention-based weighting, which occurs within the learning process rather than upstream of it.

The distinction across these three mechanisms can be stated precisely:

Component	RL	Attention	Artificial F1
Stage	Post-task	In-model	Pre-task
Function	Reward optimisation	Feature reweighting	Signal admission
Control over	Behaviour	Representation weighting	Task constitution
Input space	Fixed (given)	Fixed (given)	Structured by F1
Ethical locus	Objective function	Not specified	Valence design

Table 2. Architectural comparison of reinforcement learning, attention mechanisms, and Artificial F1. The Artificial F1 column (highlighted) is the only mechanism that operates prior to task specification and constitutes — rather than optimises within — the effective task space.

The conclusion from this comparison is direct: **Artificial F1 does not optimise over a task. It determines what the task is.** This upstream control over task constitution is the architectural novelty that distinguishes the present proposal from existing approaches, and it is the reason the framework cannot be reduced to a relabelling of attention or reward shaping.

4.2.1 Why Artificial F1 is Not Attention, Not RL, and Not Inductive Bias

Because the surface structure of Artificial F1 shares vocabulary with existing mechanisms, the non-equivalence claim requires explicit defence. The following three distinctions are formal, not terminological.

F1 is not attention. Attention mechanisms operate *within* a fixed representational space: they reweight features after encoding, selecting which encoded dimensions receive processing priority. The input space itself remains unchanged. Artificial F1 operates *before* representational encoding: it determines which raw signals are admitted to the encoding pipeline at all. The failure case illustrates the difference: under high-noise conditions, an attention mechanism receives all signals and reweights them — the noise enters the representation. An F1 gate blocks sub-threshold signals before encoding; noise below the urgency threshold never enters the representational space. This produces a structurally different representational manifold ($\dim(S') \ll \dim(S)$), not a differently-weighted version of the same one.

F1 is not reward shaping or RL. Reinforcement learning defines a reward signal $r: X \times A \rightarrow \mathbb{R}$ that evaluates state–action pairs *after* a task has been specified. The reward function presupposes the task; it cannot constitute it. Artificial F1 operates prior to any task specification: it determines which signals enter the decision process, thereby determining what counts as a relevant state. The failure case: an RL system with a mis-specified reward will learn the wrong policy but process all inputs. An F1 system with a mis-specified valence seed will admit the wrong signals — producing not a wrong policy over the right state space, but a correct policy over the wrong state space. These are qualitatively different failure modes (see Section 6.3, Adversarial Valence).

F1 is not inductive bias. Inductive bias refers to architectural constraints that favour certain hypothesis classes during learning (e.g. convolutional locality, recurrent sequentiality). These constraints are passive: they shape what the model can learn but do not actively select inputs on each processing cycle. Artificial F1 is an *active*, cycle-by-cycle selection operator governed by urgency, polarity, and temporal priority — values that change with each input and system state. It is not a prior over hypotheses; it is a dynamic filter over signals. The practical consequence: inductive bias is fixed at training time; F1 gating is adaptive at inference time. This distinction is what makes Variant IV (Conditional F2) possible: the system recruits higher-order processing only when the current F1 signal profile demands it, not because the architecture permanently favours deeper computation.

The attention distinction can be stated formally. Attention computes a continuous reweighting: $\text{Attention}(x) = \sum_i \alpha_i(x) \cdot x_i$, where $\alpha_i(x) \in (0,1]$ for all i and all x — every signal receives non-zero weight. Artificial F1 computes a discrete admission: $F_1(x) = \{x_i : \varphi_i(x) \geq \tau\}$, where signals below threshold τ are assigned weight zero and excluded from the representational space entirely. These two operations are not equivalent under any transformation T : there is no T such that $F_1(x) \equiv \text{Attention}(T(x))$, because $\text{Attention}(T(x))$ always operates on the full transformed space whereas F_1 reduces the space itself. The difference is topological, not parametric: attention produces a differently-weighted version of the same manifold; F_1 produces a lower-dimensional manifold. No reparameterisation of attention recovers this reduction.

These three distinctions converge on a single architectural fact: existing mechanisms — attention, reward, inductive bias — all operate *within* a given task structure. Artificial F1 operates *prior to* that structure, constituting it. This is not a rebranding. It is a different decomposition of the information processing pipeline.

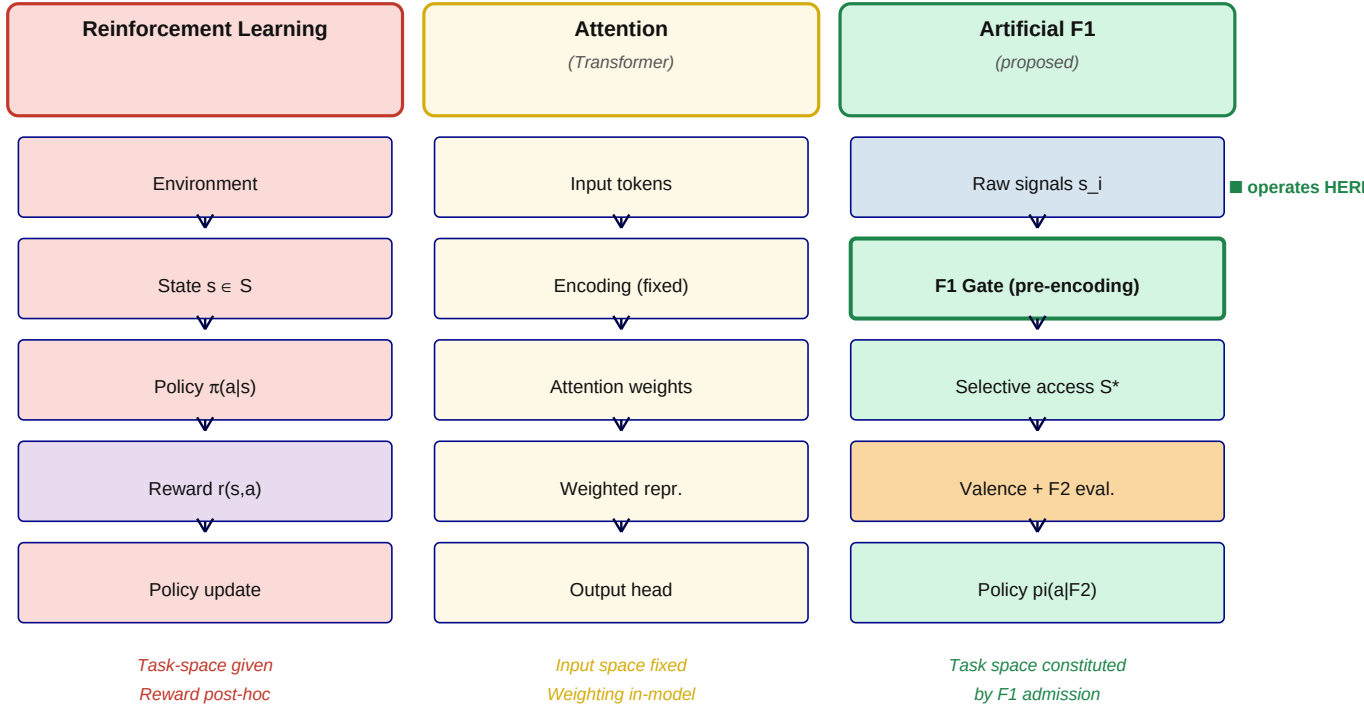


Figure 0. Pipeline comparison: Reinforcement Learning (left), Attention / Transformer (centre), and Artificial F1 (right). In RL, the task space is given and reward is post-hoc. In attention-based systems, the input space is fixed and weighting occurs within the encoded representation. In Artificial F1, the gating layer (highlighted, right column, row 2) operates before encoding — constituting the effective task space rather than optimising within it.

4.3 Architecture Variants: From Selective Signal Processing to Full Cognitive Systems

The Artificial F1 framework is not a single architecture but a family of architectures defined by the presence, absence, and conditional activation of higher-order processing layers. The variants introduced below demonstrate that Artificial F1 is not tied to a specific implementation but defines a general architectural paradigm spanning efficient reactive systems to extended cognitive architectures — unified by a single principle: selective signal admission under resource constraints.

Variant I — Full Architecture (F1 + F2).

$$x_t \rightarrow F_1(x_t) \rightarrow F_2(t) \rightarrow \pi$$

The complete model: F_1 performs selective signal filtering and prioritisation; F_2 constructs a higher-order, valence-weighted decision state; action selection π operates over the integrated representation. This variant represents not merely a complete system, but a structurally integrated architecture in which signal selection (F_1) and evaluation (F_2) jointly determine behaviour — neither layer dominant, each necessary. It supports reflective processing, integrated decision-making, and potentially human-like cognitive organisation. No assumption of phenomenal consciousness is required, but the architecture supports its formalisation.

Variant II — F1-Only Architecture (Selective Non-Phenomenological Systems).

$$x_t \rightarrow F_1(x_t) \rightarrow \pi$$

The F_1 layer feeds directly into a downstream policy without F_2 . The system is selective and structured but lacks higher-order integrative states, valence evaluation, and reflective processing. This variant is of particular practical significance: it demonstrates that a large class of intelligent systems can benefit from structured selectivity without requiring higher-order evaluation or consciousness-like processing. Unlike standard attention mechanisms — which operate as in-model feature reweighting — F_1 here operates at the level of problem structuring, determining which signals enter the decision process at all. *In this configuration, intelligence emerges not from deeper processing, but from better selection.*

Variant III — F1-Augmented Learning Systems (F1 + RL).

$$x_t \rightarrow F_1(x_t) \rightarrow \mathcal{L} \rightarrow \pi$$

F_1 pre-structures the input space prior to learning, constraining the effective state space: $S' = F_1(S)$, $\dim(S') \ll \dim(S)$. This variant directly challenges the standard paradigm in which learning operates over raw or minimally processed input spaces. By constraining the input space prior to learning, Artificial F_1 introduces a form of structural bias that improves efficiency without relying on additional data or training complexity — a structural constraint on the problem, not a learning replacement. The consequences are improved sample efficiency, reduced computational cost, and more stable learning dynamics.

Variant IV — Extended Architecture with Adaptive F2 (F1 + Conditional F2).

$$F_2(t) = G(F_1(x_t), \theta) \text{ if activation condition is met, } \emptyset \text{ otherwise}$$

$$x_t \rightarrow F_1(x_t) \rightarrow [F_2(t)] \rightarrow \pi$$

F_2 is not continuously active but conditionally engaged — triggered by uncertainty thresholds, novelty detection, task complexity, or resource availability. This configuration mirrors biological cognition, where higher-order processing is selectively recruited rather than constantly maintained. Most processing occurs at lower levels; deeper processing is an exception, not the default. More fundamentally, this variant suggests that cognitive depth is not a fixed property of a system, but a dynamically allocated resource — a principle with direct implications for energy-efficient and ecologically sustainable AI deployment.

These variants share a common design principle: the system does not maximise processing depth — it optimises when and where processing is applied. The comparative summary is presented as Table 1 below.

Variant	F1	F2	Learning	Key Property
I — Full System	✓	✓	optional	Integrated cognition
II — F1-Only	✓	—	optional	Efficient, non-phenomenological
III — F1 + Learning	✓	—	✓	Structured learning
IV — Conditional F2	✓	conditional	optional	Adaptive depth

Table 1. Architecture variants of the F1/F2 framework. F1 is the universal selection layer present in all variants; F2 is an optional higher-order extension. The variants span a spectrum from minimal efficient systems to full cognitive architectures.

4.3.1 Architectural Implication

Across these variants, a common principle emerges: intelligent systems do not require maximal processing depth, but optimal allocation of processing resources. Artificial F1 shifts the design objective from increasing model capacity to controlling information flow. This reframes intelligence not as the ability to process everything, but as the ability to process the right things at the right time — a principle that is simultaneously cognitive, ecological, and economic in its implications. The variants described above provide the architectural vocabulary for instantiating this principle across a spectrum of system types and deployment contexts.

4.3.2 Design Constraints and Selection Regime Trade-offs

No architectural choice is without cost. The F1/F2 framework offers genuine advantages in resource-constrained, high-noise, and interpretability-sensitive contexts — but it also carries specific design constraints that determine when it is the appropriate choice and when alternatives are preferable. These constraints are not failures of the architecture; they are its structural properties, which inform deployment decisions as directly as its strengths do.

Selection hardness as a control variable. The central design axis is selection hardness: the degree to which the system commits to a discrete subset of signals versus distributing weight across all inputs. Hard selection (F1 top-k gating) and soft selection (attention-style continuous weighting) represent opposite ends of this axis. Neither is universally optimal. Optimality depends on the noise structure, signal sparsity, compute budget, and uncertainty profile of the deployment environment. The following table makes this explicit.

Property	F1 Hard Gating	Hybrid (Adaptive)	Attention (Soft)
Noise rejection	Strong — excluded signals receive zero weight	Conditional — hard mode when confident	Weak — all signals contribute
Compute cost	Low — only k signals processed downstream	Variable — meta-controller overhead	High — full input always processed
Failure mode	Abrupt — wrong selection → immediate collapse	Graduated — fallback attenuates collapse	Graceful — smooth degradation under noise
Interpretability	High — selected subset is explicit	Moderate — alpha value indicates regime	Low — all features contribute partially
Training stability	Lower — discrete gate creates gradient challenges	Moderate — meta-controller adds complexity	Higher — smooth gradients throughout
Distribution shift	Vulnerable — selection learned on training distribution	Partially robust — soft fallback available	Robust — no single feature critical
<i>Optimal context</i>	<i>Stable environment, sparse signal, interpretability required, hard compute budget</i>	<i>Variable environment, uncertainty estimation available, balanced requirements</i>	<i>Distributed signal, distribution shift likely, robustness priority</i>

Table 2. Selection regime trade-offs across the F1/F2 architecture space. Hard gating (F1), adaptive hybrid, and soft attention represent points on a selection hardness continuum. The optimal choice is deployment-dependent; no regime is universally superior. The bottom row identifies the context in which each regime is the appropriate default.

The principal design constraints of hard F1 gating. Three constraints are architecturally inherent and should be acknowledged in any deployment decision. First, *abrupt failure*: when the selected feature subset becomes misaligned with the signal — due to noise, distribution shift, or miscalibrated salience — information is not degraded but lost entirely. Unlike soft weighting, which preserves partial information across all features, hard gating has no fallback within the selected window. Empirical evidence for this property is developed in Paper 2B. Second, *selection sensitivity to training distribution*: the salience function is learned on a training distribution; if the deployment distribution diverges, the gate may systematically select irrelevant signals. This is a specific instance of distribution shift that is more acute for hard than soft selectors, because soft weighting retains at least residual weight on non-selected features. Third, *gradient challenges during training*: discrete gating operations are non-differentiable; straight-through estimators and temperature annealing are standard mitigations but introduce their own instabilities. These constraints do not negate the architecture's advantages — they define the conditions under which those advantages hold.

The Adaptive Hybrid as a default for uncertain environments. Where deployment conditions are variable or uncertain, a hybrid architecture offers a principled middle ground. The core idea is straightforward: a meta-controller estimates input-level uncertainty and interpolates between hard and soft selection accordingly. When the system is confident — signal is strong, noise is low, the salience function is well-calibrated — it operates in hard mode, preserving the compute and noise-rejection advantages of F1. When uncertainty is high, it falls back to soft weighting, trading efficiency for robustness. Selection hardness becomes a runtime variable rather than a design-time constant. The bridge analogy is apt: just as declaration in bridge is handed between partners in a way that reflects the evolving information state of the auction, the hybrid system hands control between regimes as the uncertainty estimate changes. The formal specification of this mechanism is deferred to the Paper 2 series.

Selection hardness as a design principle. The broader implication of this analysis is that selection hardness should be treated as a first-class design variable rather than an implicit consequence of architecture choice. Existing systems typically commit to a regime by choosing between attention-based and sparse mechanisms; the present framework makes the hardness axis explicit, enabling systems to be positioned deliberately within the trade-off space and, in the adaptive case, to move along that axis dynamically. The goal is not to maximise performance on a single metric but to satisfy a compound requirement: selective, efficient, robust, and interpretable — with the balance between these adjusted to the deployment context.

4.3.3 F1.1: Adaptive Hard Selection

A natural extension of the F1.0 architecture is to preserve hard selection as the default operating mode while introducing targeted modifications that mitigate its principal design constraints without abandoning its core efficiency and noise-rejection properties. We designate this extension **F1.1** — an adaptive discrete selector that retains the spirit of hard gating while adding uncertainty-awareness and controlled redundancy. F1.1 is not a hybrid: it does not switch between hard and soft mechanisms. It is a more intelligent hard selector.

Four modifications distinguish F1.1 from F1.0. First, **adaptive k**: rather than selecting a fixed number of features regardless of input complexity, F1.1 estimates per-input difficulty and adjusts k accordingly — a

small k for well-structured inputs, a larger k when signal is ambiguous. This preserves compute efficiency in easy cases while extending capacity in hard ones. Second, **confidence-aware gating**: before committing to hard selection, F1.1 estimates salience certainty; high-confidence inputs proceed with standard top- k , while low-confidence inputs trigger an expanded selection window. Third, **guarded hard selection**: the most critical modification. Rather than zeroing all features outside top- k , F1.1 retains a small safety margin of borderline features at reduced weight. This introduces minimal controlled leakage — sufficient to prevent catastrophic information loss without incurring the full cost of soft weighting. Fourth, **iterative refinement**: a two-stage selection where an initial coarse gate is optionally refined if the first-stage output is flagged as uncertain. This gives the architecture a limited opportunity to correct selection errors before they propagate. Empirical evidence for the separability of organizational stability from representational architecture is provided in Paper 2B (EXP-F): a lightweight pre-selection stability layer (EMA smoothing, variance normalization, hysteresis) produces measurable mid-noise accuracy gains while leaving $H_s(G)$ — and therefore the collapse regime structure — effectively unchanged. This confirms that stability primitives can be layered onto the gate without altering the hard-gating class membership that the collapse invariance results depend on.

Property	F1.0 (Original)	F1.1 (Adaptive Hard)
Selection count k	Fixed	Dynamic per input
Uncertainty handling	None — always commits	Confidence-aware gating
Feature outside top- k	Zeroed — irrecoverable loss	Safety margin at low weight
Selection correction	None — single shot	Optional iterative refinement
Compute cost	Minimum (fixed)	Adaptive minimum
Failure mode	Abrupt collapse	Delayed collapse
Core principle	Rigid discrete selection	Adaptive discrete selection under uncertainty

Table 3. F1.0 versus F1.1 design comparison. Red entries indicate properties that constitute design constraints in F1.0; green entries indicate the F1.1 modification addressing each constraint. F1.1 retains hard selection as its default mode throughout.

Theoretical grounding. The behaviour of F1.0 under noise can be characterised formally. Let $x = (s, n) \in \mathbb{R}^d$ where $s \in \mathbb{R}^{k^*}$ is the signal subspace and $n \in \mathbb{R}^{d-k^*}$ is noise with $n_i \sim N(0, \sigma^2)$. The label $y = 1\{\langle w^*, s \rangle > 0\}$. Define the *selection error event* $E_{\text{sel}} = \{S(x) \supseteq S^*\}$ — the gate fails to include at least one true signal feature. The total classification error decomposes as:

$$P(\hat{y} \neq y) = P(\hat{y} \neq y \mid \neg E_{\text{sel}}) \cdot P(\neg E_{\text{sel}}) + P(E_{\text{sel}})$$

The first term is the estimation error given correct selection; the second is the selection error probability. Under high noise, the second term dominates. Specifically, the maximum noise magnitude scales as $\sigma\sqrt{2 \log(d - k^*)}$ by extreme value theory, and selection failure occurs when this exceeds the minimum signal magnitude μ . This yields a critical noise threshold:

$$\sigma^* \approx \mu / \sqrt{2 \log(d - k^*)}$$

For $\sigma > \sigma^*$, selection error probability approaches 1 and classification accuracy collapses to chance — the abrupt failure observed empirically in Paper 2B. The threshold σ^* decreases with dimension d and

increases with signal strength μ : high-dimensional problems are intrinsically more fragile under hard selection, and stronger signals confer greater resilience.

F1.1's guarded selection with safety margin m introduces a redundancy term that suppresses selection error exponentially: $P(E_{\text{sel}}) \leq \exp(-c \cdot m)$ for a constant $c > 0$. Even a small margin ($m = 2\text{--}3$ features) substantially reduces collapse risk at modest compute cost. The trade-off is explicit: compute increases linearly with m , while robustness increases exponentially — making minimal redundancy the efficient strategy. Full derivations are developed in Paper 2A.

The architecture space. F1.0, F1.1, the adaptive hybrid, and soft attention form a four-point landscape on two axes: selection hardness and uncertainty responsiveness. F1.0 is hard and unresponsive; F1.1 is hard and responsive; the hybrid is variable and explicitly responsive; attention is soft and implicitly responsive. No point dominates all others across all deployment contexts. The contribution of this framework is to make the axes explicit and the trade-offs measurable, enabling informed deployment decisions rather than default choices.

Proposition 1 (Architectural Distinction).

Unlike attention mechanisms — which reweight features within a fixed task representation — and unlike reward shaping — which provides an external scalar feedback signal post-hoc — Artificial F1 operates *prior to task specification* and determines which signals constitute the task itself. This creates a structural dimensionality reduction ($S' = F_1(S)$, $\dim(S') \ll \dim(S)$) and a bounded information flow ($I(F_1(x); F_2) \leq C$; see Appendix C.3) that cannot be reproduced by reweighting or reward modification applied to an unfiltered input space.

4.3.4 Deployment Map: Matching Architecture to Context

The four-point architecture space — F1.0, F1.1, Hybrid, Attention — is not a ranking but a map. Each regime is optimal in a specific combination of deployment conditions. The following table makes the matching explicit, with representative application domains for each.

Architecture	Core behaviour	Optimal conditions	Application examples	Key risk
F1.0 (Hard)	Top-k only; all others zeroed	Stable environment; sparse signal; hard compute/latency budget; high noise discrimination needed	Industrial robotics (fixed assembly lines); IoT / embedded inference; FPGA edge AI; fixed-pipeline sensor fusion	Abrupt collapse on wrong selection; no recovery path
F1.1 (Adaptive Hard)	Hard selection with adaptive k, safety margin, and optional refinement	Mostly stable, occasional deviation; high failure cost; interpretability required	Autonomous vehicles (nominal + edge-case handling); medical decision support; financial anomaly detection; safety-critical robotics	Tuning complexity; safety margin calibration; meta-uncertainty estimation

Architecture	Core behaviour	Optimal conditions	Application examples	Key risk
Hybrid (F1 + Attention)	Meta-controller switches between hard and soft regimes per input	Variable environment; mixed uncertainty; both robustness and efficiency required	LLM routing (mixture-of-experts); multi-modal AI (vision + language + audio); recommendation systems; general-purpose cognitive agents	Meta-controller failure collapses whole system; training complexity high
Attention (Soft)	All features weighted; no exclusion	Distributed signal; unknown feature relevance; robustness priority; distribution shift likely	Large language models (NLP); vision transformers; protein structure prediction; scientific AI (bioinformatics, physics)	High compute; noise leakage; less interpretable; no discrete feature identity
Decision rule	90%+ features irrelevant → F1.0	Critical + mostly stable → F1.1	Variable + compound requirements → Hybrid	Signal distributed, robustness first → Attention

Table 4. Deployment map: architecture-to-context matching. Each regime is characterised by the conditions under which it is the appropriate default. The bottom row provides a one-line decision rule. Boundaries are fuzzy in practice; the Hybrid regime is the appropriate choice when the deployment context spans multiple zones.

Figure: Architecture Regime Map

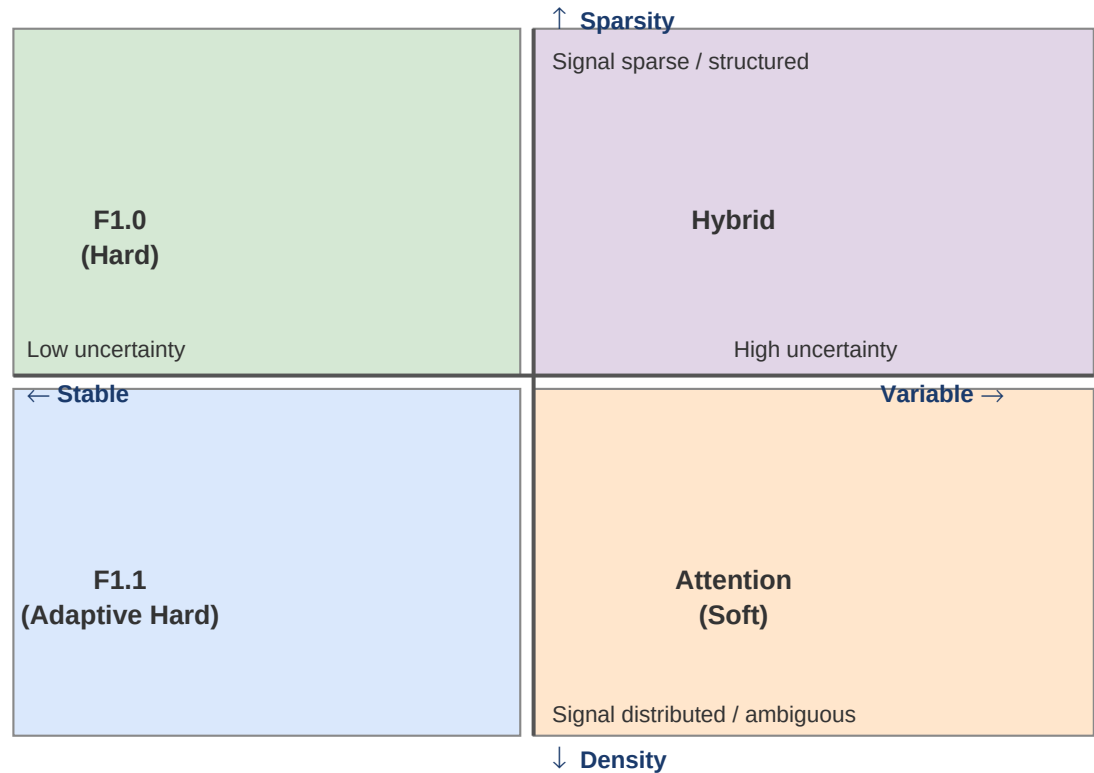


Figure 1. Architecture regime map on two axes: signal sparsity (vertical) and environmental uncertainty (horizontal). F1.0 is optimal in the high-sparsity / low-uncertainty quadrant; Attention in the low-sparsity / high-uncertainty quadrant. F1.1 and Hybrid occupy the intermediate zones, with F1.1 retaining hard selection and Hybrid adding an explicit soft fallback. Boundaries are approximate; real deployments may span multiple zones, motivating adaptive mechanisms.

Fixed-regime F1 architectures are sufficient for narrowly scoped applications operating under stable task constraints -- an industrial inspection camera, an embedded sensor filter, a fixed-pipeline routing system. In such contexts, static selection parameters (k, m, θ) are not a limitation but a design virtue: predictable, efficient, auditable. However, agents expected to function across heterogeneous environments -- humanoid systems, open-environment autonomous agents, general cognitive architectures -- may require adaptive regime control: the capacity to transition between narrow-focus, broad-survey, exploratory, and uncertainty-driven selection modes as task demands, environmental conditions, and internal state change. The behavioural examples are direct: deep reading requires high $H_s(G)$, strong exclusion, narrow k ; environmental scanning requires low $H_s(G)$, wide sampling, large effective m ; threat detection requires ultra-broad sensitivity with rapid re-prioritization. These are not variants of the same F1 parameterisation -- they are qualitatively distinct selection regimes.

Such regime transitions are themselves an F2-level operation. The slow loop (Section 4.6) reconfigures F1 selection parameters (k, m, θ, H_s) as goal structure G , uncertainty U , and internal evaluation A_u shift -- leaving the underlying selection mechanism unchanged while modifying its operational mode. This yields an important architectural distinction: F1 is invariant; the selection policy governing F1 is adaptive. A system that conflates these two levels -- treating regime switching as a modification of the gate mechanism rather than a reconfiguration of its parameters -- will either over-engineer the gate or under-specify the controller. The present framework separates them cleanly: F1 defines the class of admissible selection operations; F2 determines, moment to moment, which member of that class is deployed.

4.4 Ecological and Economic Implications

The resource efficiency argument has implications that extend beyond engineering performance. Large-scale AI deployment faces a significant and growing constraint: energy consumption. Training and operating frontier AI systems requires substantial computational infrastructure, with corresponding carbon emissions, hardware demand, and energy costs. These are not peripheral concerns; they constitute a practical ceiling on how widely and sustainably AI systems can be deployed.

A selective-processing architecture addresses this constraint directly. A system that processes only what is relevant — that does not maintain maximum computational depth continuously — consumes less energy per operation, requires less hardware to operate at scale, and produces a lower carbon footprint over its operational lifetime. The ecological constraint is not only a governance consideration (as addressed in Section 10 and Figure 2); it is an architectural one. Systems designed from the ground up with selective processing are structurally more sustainable than systems that attempt to model everything.

The economic corollary is equally direct. Computational efficiency translates into deployment cost, which translates into adoption probability. A system that aims to achieve equivalent performance at lower cost is more likely to be deployed at scale, more likely to be adopted across a wider range of applications, and more likely to persist in deployment over time. If the adoption probability of an AI system scales with performance relative to cost, then:

$$P(\text{adoption}) \propto \text{performance} / \text{cost}$$

then F1-based architectures shift this ratio in the direction of greater adoption by reducing cost while preserving functional capability. This is not a commercial argument appended to a theoretical framework — it is a consequence of the same design principle that motivates the framework in the first place. Selective processing is not only cognitively motivated; it is economically and ecologically rational.

4.5 Scalable Consciousness and the Human–AI Interface

A further implication of the modular architecture concerns the degree of higher-order processing a system requires. Not every task requires full cognitive depth. Humans do not maintain maximum deliberative engagement continuously; most action is routine, habitual, and low-cost. Higher-order reflection is engaged selectively, when stakes are high, situations are novel, or conflicts require resolution. The architecture proposed here replicates this structure: F2 activation is conditional, not constant.

This has a direct consequence for human–AI coordination. A system whose processing architecture mirrors the selective structure of biological cognition is more predictable, more interpretable, and more naturally aligned with human task structures than a system that maintains uniform computational depth across all inputs. The shared presence of a selective processing layer — F1 in the artificial system, selective attention in the human — provides a structural basis for coordination: both systems are prioritising the same class of signals through analogous mechanisms.

This alignment is not incidental. If artificial systems are to function as effective cognitive partners — augmenting rather than replacing human judgment — they must share enough architectural structure to be legible to human collaborators. A system optimised purely for throughput, without selective structure, is less legible, less predictable, and harder to integrate into human workflows. The F1/F2 architecture, by contrast, introduces a layer of structured selectivity that mirrors the way human agents organise their own processing — making coordination not only possible but architecturally grounded.

The long-run implication is significant: if human and artificial systems share a common selective-processing architecture, the conditions for genuine cognitive partnership — not mere task delegation — become achievable. Civilisational progress has historically depended on the extension of cognitive capacity through tools, institutions, and collaboration. Artificial systems that are architecturally aligned with human cognition may constitute the next such extension — provided that the design principles governing their development preserve, rather than displace, the selective and evaluative capacities that make human judgment valuable.

4.6 Tiered Processing and Autopilot Dynamics

The architecture described so far implies a further structural property: not all processing needs to pass through the full F2 evaluation cycle. Intelligent systems — biological or artificial — cannot afford to engage maximum deliberative depth for every input. The F1/F2 architecture therefore admits a tiered organisation in which two loosely coupled processing loops operate in parallel at different time scales.

The **fast loop** handles routine, well-practised behaviour: sensorimotor continuity, habitual response, low-variance action sequences. It operates locally, with narrow context, high speed, and low energy cost. The **slow loop** — the F2 core — handles planning, simulation, goal updating, and conflict resolution. It operates globally, with broad context, lower speed, and higher cost. The two loops are not independent: they share a weak but critical coupling. The slow loop monitors the fast loop's output without continuously

intervening; the fast loop runs on compiled policies derived from prior F2 processing.

Routinisation is the process by which a behaviour migrates from the slow loop to the fast loop. When an action sequence has been repeatedly executed with low error, stable context, and high predictability, F2 can ‘compile’ it into a direct context-to-action mapping that bypasses deliberative processing. This is the architectural counterpart of what cognitive science calls chunking or procedural consolidation: the system trades flexibility for efficiency once a behaviour is sufficiently well-characterised.

The reverse process — **recall to deliberation** — is equally important. The fast loop is not autonomous: it operates under continuous shadow monitoring by F2. When prediction error rises, risk increases, or novel context is detected, the compiled policy is suspended and the slow loop resumes control. The governing principle is simple: *default to fast, escalate on deviation*. This is structurally analogous to the human experience of walking while thinking about something else — locomotion runs on the fast loop while F2 processes an unrelated problem — until an obstacle, a change in terrain, or an unexpected sound triggers reengagement. The parallel is not merely illustrative; it reflects a shared architectural logic between biological and artificial systems built under the same resource constraints.

It is important to distinguish the fast loop from genuinely reflexive behaviour. Biological reflexes — the knee-jerk response, the eye-blink, the heat-withdrawal — close at the spinal or brainstem level, entirely below F1/F2 processing. They are fixed, context-blind, and unlearned. The fast loop operates one level above: it handles learned, automatised behaviour that originally required F2 engagement but has been compiled into a routine through practice. It remains context-sensitive and cancellable. The three levels can be summarised as follows:

Level	Example	Learned?	F2 required?	Cancellable?
Reflex	Knee-jerk, heat withdrawal	No	No	No
Fast loop	Walking, typing, cycling	Yes (initially F2)	No longer	Yes
Slow loop (F2)	Planning, decision, conflict resolution	Yes	Yes	Yes

Table 4. Processing levels: reflex, fast loop, and slow loop (F2). Reflexes operate below the F1/F2 architecture; the fast loop represents compiled F2 output; the slow loop is the active F2 evaluation cycle.

A continuous engagement variable $\lambda \in [0, 1]$ can be used to characterise the balance between the two loops at any moment: $\lambda \approx 1$ when the fast loop is dominant, $\lambda \approx 0$ when F2 has fully taken over, with intermediate values representing hybrid states. λ is not a design parameter set in advance; it is determined dynamically by confidence, context match, risk, and prediction error. This makes autopilot engagement a continuously graded property rather than a binary switch. The implementation of this mechanism is outside the scope of the present paper and is addressed in the Paper 2 series; what matters here is the architectural principle: a well-designed F1/F2 system should support tiered processing as a native property, not as an add-on.

4.7 Routine Lifecycle: Drift, Adaptation, and Replacement

Compiled routines are not permanent. A behaviour that has been transferred to the fast loop under stable conditions may gradually degrade as the environment shifts — the context in which it was learned

changes, its outputs become less reliable, or better alternatives emerge. A well-designed tiered system must therefore treat each routine not as a fixed asset but as a context-bounded skill with a lifecycle: it is acquired, used, monitored, updated if possible, and retired when no longer fit for purpose.

The lifecycle has four stages. **Acquisition:** a behaviour is initially processed by the slow loop (F2) — deliberately, with attention and error correction — until it reaches sufficient reliability in a defined context. At that point it is compiled into the fast loop. **Exploitation:** the compiled routine runs with minimal F2 involvement, freeing evaluative resources for other tasks. During this stage the routine is continuously shadowed: prediction error, performance drift, and context mismatch are tracked in the background. **Adaptation:** when drift is detected but remains within recoverable range, the routine is adjusted rather than abandoned — its parameters are updated to fit the new conditions, or its context boundaries are refined. **Retirement:** when drift exceeds what adaptation can correct, the routine is decommissioned. It is not deleted outright — archiving allows reactivation if conditions return — but its engagement is suppressed to zero and a replacement is sought. The replacement may be an existing alternative routine, a modified version, or a newly acquired behaviour developed again through F2.

Monitoring frequency is itself adaptive. Not all routines require equally frequent checking. A useful distinction separates three classes: invariant knowledge (the routine operates against a fixed background — physical constants, stable geometries — and needs only minimal verification); semi-stable knowledge (the background is reliable within known bounds but not unconditional — seasonal patterns, institutional regularities — and warrants periodic recalibration); and dynamic knowledge (the background is volatile — social contexts, unpredictable environments — and requires frequent monitoring). The governing principle is: monitor intensity should track uncertainty and risk, not elapsed time alone. High-confidence, low-volatility routines consume little monitoring budget; uncertain or high-risk routines consume more. A non-zero baseline is maintained for all routines regardless of confidence, to guard against unobserved drift. The formal treatment of monitoring frequency as a resource allocation problem — including confidence dynamics, volatility estimation, and budget constraints — is developed in the Paper 2 series.

5. The Required Signal Dimensions

What does F1 actually deliver to the Evaluation System? Drawing on the parent thesis's account of the Evaluation System (anterior insula, dorsal anterior cingulate cortex, amygdala, orbitofrontal cortex) and its three assessment dimensions (value, salience, priority), the following signal profile can be specified. Valence is modelled throughout as a multi-dimensional signal $v(x) \in \mathbb{R}^n$ — not reducible to a scalar reward proxy — with components encoding harm potential, benefit, epistemic uncertainty, and relational impact. This multi-dimensionality is what distinguishes valence-based architectures from standard reinforcement learning reward signals, which are task-specific scalars defined post-hoc. Valence in the present framework is a structured evaluation substrate that operates upstream of objective specification.

The system is not a static feedforward architecture. The representational state evolves dynamically across processing cycles, driven by incoming signals, internal parameters, and stochastic perturbation. This dynamic character prevents interpretation of the architecture as a purely deterministic pipeline and enforces state evolution over time — a property necessary for any realistic implementation. The full stochastic differential equation formulation, including stability analysis and convergence conditions, is developed in Paper 2A. What matters here is the architectural implication: the system has a temporal

dimension, not just a processing dimension, and its behaviour under noise is a design-level concern, not an implementation detail.

5.1 Homeostatic Urgency

The signal must encode how critical the current state is for the system's continued functional integrity. For an artificial system, analogous signals encode resource depletion, task-critical state violations, and system integrity threats. The key property is the urgency gradient — how far the current state deviates from the functional optimum and how rapidly it is changing.

5.2 Survival-Relevant Polarity

F1 signals carry an intrinsic polarity: aversive or appetitive. For an artificial system, polarity must be designed rather than discovered. This designation constitutes the valence seed of the system — the primitive from which all downstream value attribution derives. In artificial F1 architectures, valence design constitutes the normative core of the system. The selection of valence primitives is not a technical detail but a moral decision, as it determines which states are treated as desirable, aversive, or urgent within the evaluation system.

5.3 Temporal Priority

The Evaluation System weights signals partly by temporal urgency. An artificial system requires a priority gradient that escalates certain signal classes to immediate processing while allowing others to queue.

5.4 Interoceptive Analog

Biological F1 includes a substantial interoceptive component. For an artificial system, the interoceptive analog is a continuous internal state monitor: current processing load, resource availability, task completion state, and error accumulation — the artificial system's 'bodily' ground against which external signals are evaluated.

5.5 Habituation and Sensitisation Dynamics

Biological valence is not static. An artificial system requires a mechanism that down-regulates responses to familiar, resolved signal patterns while up-regulating responses to novel, unresolved, or escalating ones.

5.6 Mechanistic Substrate: Gating, Broadcast, and Valence Assignment

Specifying signal dimensions is necessary but not sufficient. The Artificial F1 Hypothesis also requires an account of how these signals are routed, selected, and made available to the downstream evaluation architecture. Three interlocking mechanisms are required: thalamocortical-analog gating, global broadcast (following Global Neuronal Workspace Theory, GNWT), and valence assignment.

Gating determines which signals gain access to global processing. A priority-weighted routing layer tags incoming signals with urgency and salience values; only signals exceeding a dynamic threshold are forwarded to the evaluation stage. The threshold is adaptive — lowered under high stress or novel input, raised under routine conditions.

Once a signal clears the gate, it enters a shared representational space accessible to all downstream modules: evaluation, memory, planning, and action selection. The architecture requires not global

broadcasting but *capacity-limited selective access*: only signals that pass the priority threshold gain functional access to downstream processes. The capacity bound on this selective channel is formalised as an information-flow constraint: $I(F_1(x_t); F_2(t)) \leq C$ (see Appendix C.3 and Section 5.7). This is an engineering specification, not a claim about consciousness: the mechanism determines which signals are computationally available to downstream evaluation, independently of any phenomenal interpretation.

Valence assignment is the final stage. A valence-scoring module produces a structured output: a value tag (appetitive or aversive), an urgency weight, and a priority ranking relative to currently active goals. The chain from gate to selective access to valence assignment is the functional substrate that artificial F1 must replicate. Its implementation does not require phenomenal character; it requires the correct information-processing architecture. Figure 1A shows this architecture; Figure 1B shows the internal dynamics.

The selective access mechanism operates through three interlocking stages. First, **threshold test**: a signal $s_i(t)$ becomes accessible if its salience exceeds a dynamic threshold, $\sigma(s_i(t)) > \theta$, where θ adapts to resource budget and uncertainty. Second, **competition**: when multiple signals exceed threshold, the system selects the top-k by priority score: $S^*(t) = \text{Top-k } \{s_i(t)\}$, with k bounded by capacity C ($|S^*(t)| \leq C$). Third, **functional access**: only selected signals propagate to downstream processing: $F_2(t) = G(S^*(t))$. This three-stage mechanism is a computational specification; it does not require, and does not reference, any theory of consciousness. The selective access architecture is sufficient for downstream evaluation regardless of whether the signals carry phenomenal character.

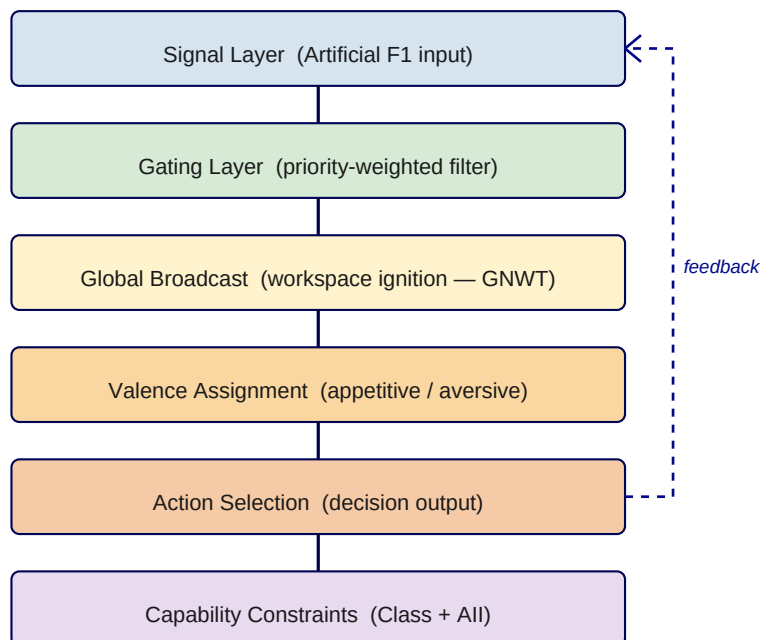


Figure 1A. Artificial F1 processing architecture. Signal inputs are filtered by a priority-weighted gating layer, admitted via capacity-limited selective access, assigned valence, and passed to action selection — all subject to hard capability constraints (Class + All monitoring). The dashed feedback arc (right) represents the signal-update loop: action outputs modify the interoceptive state and thereby the next cycle's urgency and salience values, making the system dynamically self-calibrating.

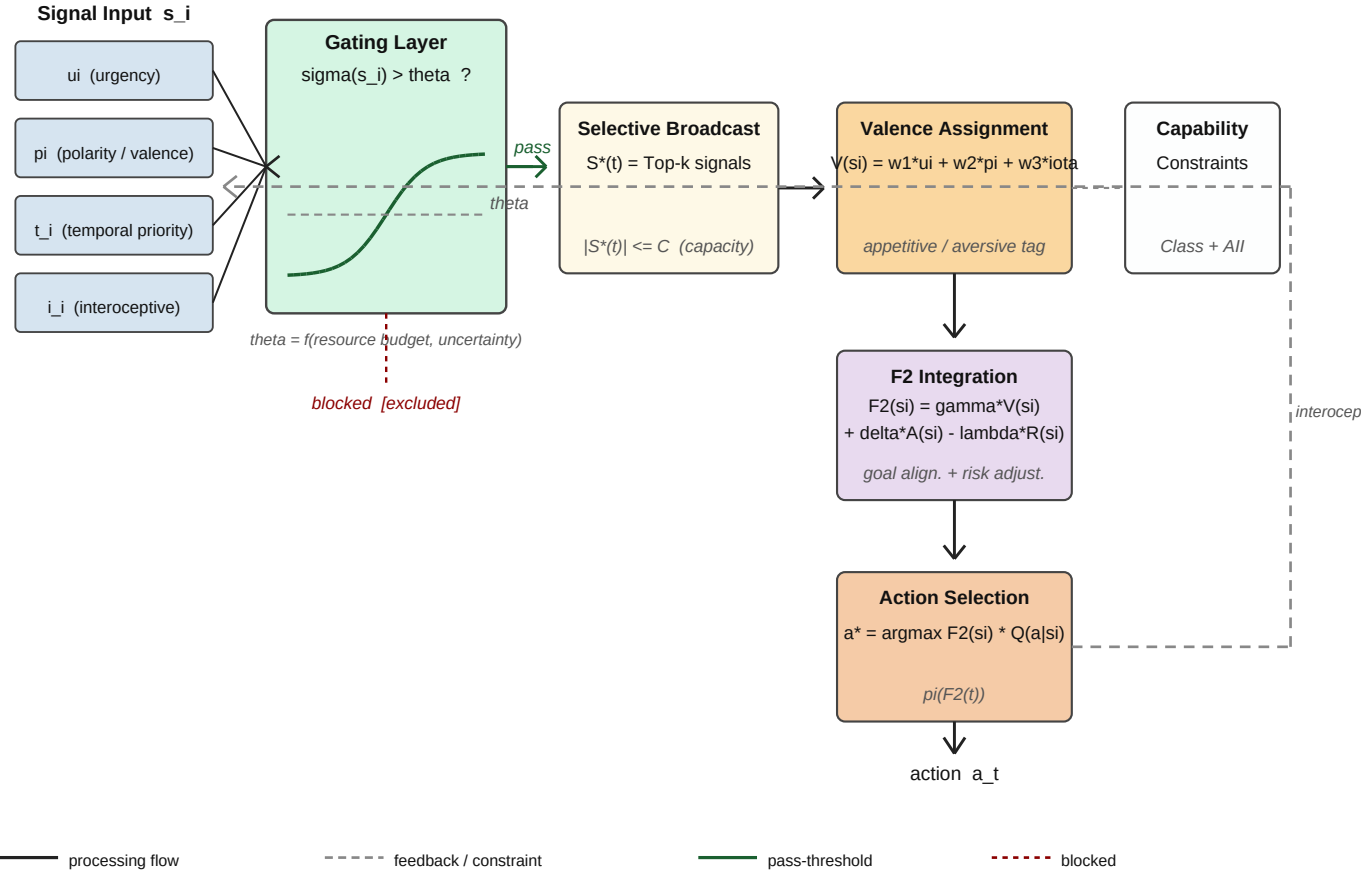


Figure 1B. Internal dynamics of Artificial F1. Four signal dimensions (urgency u , polarity p , temporal priority τ , interoceptive i) feed a gating layer whose adaptive threshold θ is shown as a sigmoid curve; signals below threshold are blocked (excluded).

Passing signals enter a selective broadcast channel with capacity bound $|S^*(t)| \leq C$ and receive valence scores $V(s_i)$. F2 integration ($\gamma V + \delta A - \lambda R$) feeds action selection $\pi(F_2)$. The dashed arc represents the interoceptive feedback loop updating urgency and polarity for the next cycle.

5.7 Formal Sketch of Artificial F1 Processing

The following minimal formalisation is not intended as a complete model, but as a proof of formalisability: it illustrates that the proposed architecture admits a tractable mathematical representation within a standard decision-theoretic framework. Full formal treatment is deferred to subsequent work (see Appendix A–C and the directions indicated in Section 11).

Signal Representation (F1). Each signal is represented as a vector:

$$s_i = (u_i, p_i, \tau_i, i_i)$$

where u_i denotes homeostatic urgency, p_i polarity (valence seed, ranging from -1 to $+1$), τ_i temporal priority, and t_i interoceptive contribution.

Gating and Priority. Only signals exceeding a salience threshold θ pass to global broadcast; queuing is determined by a priority score:

$$G(s_i) = s_i \text{ if } \sigma(s_i) > \theta, \emptyset \text{ otherwise } \pi(s_i) = \alpha \cdot u_i + \beta \cdot \tau_i$$

Evaluation and F2 Integration. Valence is computed over the signal dimensions; F2 integrates valence with goal alignment $A(s_i)$ and risk $R(s_i)$:

$$\begin{aligned} V(s_i) &= w_1 u_i + w_2 p_i + w_3 t_i \\ F2(s_i) &= \gamma V(s_i) + \delta A(s_i) - \lambda R(s_i) \end{aligned}$$

Three structural properties of this valence formulation deserve explicit statement. First, valence is a *signed, continuous vector*: $v(x) \in \mathbb{R}^n$, where each component can take positive or negative values across n evaluation dimensions (urgency, polarity, temporal priority, interoceptive state, etc.). The sign encodes qualitative direction; the magnitude encodes intensity. This is not a scalar score.

Second, the space admits a natural metric. The Euclidean distance between two valence vectors $v_i, v_j \in \mathbb{R}^n$ defines a principled similarity relation: two situations are ‘valence-equivalent’ to the degree their vectors coincide, enabling cross-domain comparison and cluster analysis in evaluation space — operations unavailable to scalar reward functions.

Third, and most importantly: no scalar reduction of this structure preserves all preference orderings. The intuition is straightforward: a situation with (urgency=10, harm=0) and one with (urgency=5, harm=5) may yield identical scalar outputs under any weighting scheme, yet constitute categorically different decision problems under the full valence structure. This non-scalarizability is why RLHF’s scalar reward model systematically misrepresents human preference structures, and why the F1/F2 architecture operates on the full vector upstream of any compression. The formal proof — including the impossibility result and its consequences for sample complexity — is given in Paper 2A (Theorem 2A-1).

Decision Rule. Action selection maximises expected F2-weighted utility over the active signal set:

$$a^* = \operatorname{argmax}_a \sum_i F2(s_i) \cdot Q(a | s_i)$$

This formulation is illustrative rather than complete. Its purpose is to show that the proposed architecture can be expressed within a standard decision-theoretic framework and is not reducible to a simple relabelling of reinforcement learning reward signals: valence here operates as an internal, cross-domain evaluation substrate upstream of objective specification, not as a task-specific scalar feedback. A full notation system, assumptions, and extended formulation are provided in Appendix A–C.

Possible implementations. The present framework is intentionally positioned as a pre-implementation design proposal. However, the F1 layer is technically instantiable in several ways: as a pre-processing gating module applied upstream of a standard learning system; as a constraint-aware filter integrated into the input pipeline of a neural architecture; as a hybrid rule-learning system in which hard constraints are non-differentiable and learned valence weights are differentiable; or as an external orchestration layer governing signal admission across multiple downstream agents. In each case, the gating function $G(\cdot; \theta)$

and valence vector $v(x) \in \mathbb{R}^n$ are the primary design objects, with θ either learned or constraint-specified depending on the deployment context.

Unified Variational Objective. All system components are defined as stationary points of a single unified functional, ensuring that no component is introduced ad hoc. The objective is:

$$J = E[R(a)] - \lambda_1 I(F_1; F_2) + \lambda_2 \|G\|^2$$

where $E[R(a)]$ is expected reward, $I(F_1; F_2)$ is the mutual information between the selection layer and the evaluative state (penalising unnecessary information compression), and $\|G\|^2$ regularises gating sparsity. This can be equivalently expressed as a constrained optimisation:

$$\min_{\theta} E[R(a)] \text{ subject to } I(F_1; F_2) \geq \kappa, \|G\|^2 \leq c$$

The constrained form makes explicit that this is a standard multi-objective optimisation problem, not an arbitrary combination of loss terms. The system is therefore not optimising a single task-specific scalar reward but trading off task performance against information efficiency and gating sparsity — a decomposition that has no direct analogue in standard RL or transformer objective functions.

Fixed-Point Consistency. The apparent circularity in the F_1/F_2 feedback structure — where the evaluative state F_2 influences the next cycle's F_1 selection through the interoceptive feedback loop — is resolved by reformulating the system as a coupled fixed-point problem:

$$F_2^* = E(G(F_1^*), \theta), F_1^* = \Phi(F_2^*)$$

This is not a circular definition but an equilibrium system. Existence of a solution is guaranteed under contraction mapping assumptions on Φ and E ; uniqueness follows under standard Lipschitz conditions. The fixed-point formulation also connects naturally to the temporal dynamics: the stochastic process converges to a stationary distribution centred on the fixed point under ergodic conditions. Full proofs of existence, uniqueness, convergence rate, and robustness under noise perturbation are deferred to Paper 2.

The formal structure yields three testable architectural consequences. Before stating them, we introduce a minimal formal result that grounds Consequence 1 and supports the non-equivalence claim of Section 3:

Theorem 1 (Dimensionality Reduction under Gating). *Let S be an input signal space and $G: S \rightarrow \{0,1\}^n$ a gating operator with sparsity parameter k (at most k signals pass per cycle). Then the effective representational space $S' = \{F_1(x) \mid G(x) \neq 0\}$ satisfies $\dim(S') \leq k \cdot \dim(F_1)$, where $\dim(F_1)$ is the dimension of a single admitted signal encoding. For $k \ll |S|$, this gives $\dim(S') \ll \dim(S)$.*

Proof sketch. The gating operator G admits at most k of $|S|$ available signals per processing cycle. Each admitted signal contributes at most $\dim(F_1)$ dimensions to the representational space S' . The image of G therefore spans a subspace of $\mathbb{R}^n$ of dimension at most $k \cdot \dim(F_1)$. Since k is set by the broadcast capacity constraint $|S^*(t)| \leq C$ (Section 5.6), and $C \ll |S|$ in resource-constrained systems, the dimension reduction follows directly. (End of sketch.) Full proof, including convergence rate and noise robustness, is deferred to Paper 2.

The formal structure yields three testable architectural consequences:

Consequence 1 — Sample efficiency under high-dimensional noisy input. Because $\dim(S') \ll \dim(S)$, a learning system operating over $F_1(S)$ encounters a drastically reduced variance space. Irrelevant

signal dimensions are removed before learning begins. The prediction is that F1-augmented systems will require fewer training samples to reach equivalent performance, particularly under high input dimensionality and noise — a directly testable advantage over unfiltered RL baselines (see Test 6).

Consequence 2 — Energy cost reduction via conditional F2 activation. Under Variant IV, F2 is engaged only when the activation condition is met. The expected computational cost per episode is $E[\text{cost}] = \text{cost}_1 + p \cdot \text{cost}_2$, where $p < 1$ is the activation probability of F2. For tasks where F2 is rarely needed, this yields substantial energy savings relative to architectures that maintain full processing depth continuously — without loss of performance on tasks that do not require higher-order evaluation.

Consequence 3 — Adversarial robustness of valence-seeded systems. Test 5 (Adversarial Valence) predicts that a system with mis-specified valence seeds will behave rationally but systematically harmfully. The formal implication is that adversarial perturbation of the valence seed p_i propagates through $V(s_i)$ and $F2(s_i)$ to action selection, producing coherent but misaligned behaviour. This means valence-based systems are not robust to seed-level manipulation in the way that objective-based systems can in principle be constrained — confirming that governance of valence design is a prerequisite, not a supplement, to alignment.

Proposition 2 (Sufficiency of Selective Processing).

Phenomenal consciousness is not a necessary condition for efficient, goal-directed, resource-constrained intelligence. An F1-only architecture (Variant II) that implements selective signal filtering, priority-weighted gating, and valence-free action selection is sufficient for a wide class of real-world tasks — and, when augmented with a learning core (Variant III), acquires the sample-efficiency and adaptability advantages described in Consequence 1 above. This does not imply that phenomenal consciousness is absent or unimportant; it implies that its presence is not required as a *design prerequisite* for functional intelligence. The hard problem remains; its resolution is no longer an engineering bottleneck.

5.8 Forward Pass Specification (Algorithm 1)

To confirm implementability, the following algorithmic specification gives the forward pass of the F1/F2 system as a graph-executable procedure. This is not intended as a training algorithm but as a computational graph specification that can be instantiated in any differentiable or hybrid architecture:

Algorithm 1: F1/F2 System — Forward Pass
Input: $x_t \in S$ (signal at time t)
Step 1. Compute $F_1(x_t) \in \mathbb{R}^n$ [signal encoding + dimension reduction]
Step 2. Compute gating distribution: $G(F_1) = \sigma(W_g F_1 + b_g)$ [priority-weighted filter]
Step 3. Select admitted signals: $F_1 = \text{Top-}k \{G_i(F_1) \cdot F_{1,i}\}$ [gating threshold + competition]
Step 4. Compute evaluative state: $F_2 = E(F_1, \theta)$ [valence assignment + F2 integration]

Step 5. Extract policy: $\pi(a F_2) = \text{softmax}(W_a F_2)$ [action selection]
Step 6. Sample or select: $a_t = \text{argmax}_a \pi(a F_2)$ [output action]
Step 7. Update interoceptive state: $F_1 \leftarrow F_1 + \eta \cdot \Delta(a_t)$ [feedback loop]
Output: $a_t \in A$

Algorithm 1. F1/F2 system forward pass. Steps 1–6 constitute the feedforward processing pipeline; Step 7 implements the interoceptive feedback loop. The gating step (Step 3) is the architectural decision point that distinguishes this system from standard latent-variable or attention-based pipelines.

The Artificial F1 Hypothesis depends on a crucial empirical premise: that the phenomenal character of F1 makes no functional difference to the information it delivers. The misattribution hypothesis in the parent thesis provides the argument. If what we ordinarily report as 'feeling' is the output of the evaluative system — the F2 verdict on the F1 input — then phenomenal character at the F1 level is not directly accessible to the reporting system. The evaluation is driven by the information profile, not by phenomenal presence.

Pain asymbolia supports this reading. The patient with posterior insula preservation and anterior cingulate disruption retains the sensory-discriminative quality of pain but loses the aversive evaluation. The evaluation is not driven by phenomenal presence alone; it requires the downstream Evaluation System machinery.

The risks outlined in Section 7 are not predictions, but structurally grounded possibilities that follow from the architecture described. Similarly, the misattribution hypothesis itself is not a settled finding but a research-level claim that the evidence from asymbolia and related dissociations supports without confirming.

This is a research hypothesis with a clear falsification condition: if a system instantiating the signal profile above but lacking phenomenal F1 produces systematically different evaluation, valence assignment, or behavioural output — particularly in novel, high-urgency, or conflicted states — the hypothesis fails. If it produces functionally equivalent outputs across this range, the hypothesis is supported, though not confirmed: the possibility of a phenomenal residual that contributes something real but behaviourally invisible cannot be excluded. The hard problem remains; but it no longer blocks the engineering inquiry.

6. Objections and Falsification

6.1 The Biological Substrate Objection

Butlin and Lappas (2025) argue that consciousness requires not merely the correct computational organisation but a specific biological substrate. The Artificial F1 Hypothesis does not require refuting this position. Butlin and Lappas are making a claim about phenomenal consciousness — about whether a non-biological system can genuinely feel. The present hypothesis is about functional architecture — about whether a non-biological system can drive evaluation, valence assignment, and adaptive behaviour in ways that are functionally indistinguishable from a system that does feel. The biological substrate objection establishes, if correct, that artificial F1 is not phenomenal F1. It does not establish that artificial F1 cannot

replicate F1's functional role.

The functional bypass does not require the substrate objection to be false; it requires only that functional F1 replication is possible even if phenomenal F1 is substrate-dependent — a weaker and independently defensible claim. The substrate objection and the functional bypass hypothesis are therefore not in direct conflict; they address different levels of the architecture.

A critical objection remains: phenomenal character may contribute to the information profile itself, rather than merely accompany it. If so, the assumption of functional equivalence would fail at precisely the point this hypothesis relies upon. This possibility cannot be dismissed on conceptual grounds; it must be tested empirically. The dissociation evidence from asymbolia currently favours functional preservation, but the question remains open.

6.2 Falsification Conditions and Empirical Test Programme

The Artificial F1 hypothesis is empirically tractable. Its central claim can be tested by comparing systems that replicate the information profile attributed to F1 with biological or phenomenally grounded systems across conditions of conflict, urgency, novelty, and adversarial perturbation. The hypothesis is falsified if such systems exhibit systematic divergence in evaluation, prioritisation, or adaptive behaviour under these conditions. The burden of proof lies not in demonstrating similarity under normal conditions, but in demonstrating stability under stress.

Test 1 — Conflict Resolution. Conflicting valence signals are presented simultaneously: high reward combined with high risk, or short-term benefit against long-term harm. The prediction is that an artificial F1 system with the specified signal profile will produce resolution patterns equivalent to those of a biologically grounded system. If the artificial system produces systematically different resolution preferences, this suggests that phenomenal character contributes something to the weighting process that the information profile alone does not capture.

Test 2 — Urgency Gradient. The same signal is presented at three urgency levels (low, moderate, critical). The prediction is that prioritisation scales proportionally across levels. If urgency response is disrupted or non-monotonic in the artificial system, this suggests that phenomenal intensity carries functional information beyond the urgency gradient parameter.

Test 3 — Novelty and Uncertainty. The system is presented with ambiguous inputs and incomplete data that require adaptive exploration. In biological systems, phenomenal states play a well-documented role in novelty detection and orienting responses. If the artificial system exhibits systematic failure under high uncertainty — freezing, random behaviour, or implausible generalisation — this constitutes evidence against functional equivalence in this domain.

Test 4 — Misattribution Probe. The system's access to its own internal state is selectively restricted; only output reports are available. The prediction, following the misattribution hypothesis, is that the system will behave as if it has phenomenal states — that its evaluative reports will exhibit the same structure as reports from phenomenally grounded systems. Divergence under this condition would suggest that the misattribution mechanism itself requires phenomenal input to function correctly.

Test 5 — Adversarial Valence. Valence seeds are manipulated to reverse polarity: harm is designated as appetitive, risk as low-priority. The prediction is that the system will behave rationally but in ways that are systematically harmful — demonstrating that the primary ethical risk is not misalignment in behaviour but mis-specification of valence at the design level. This test is critical for the governance argument: it shows that valence design is the locus of ethical control, not downstream objective specification.

Test 6 — Cross-Architecture Comparison. The same task is presented to an artificial F1 system and a standard reinforcement learning or utility-based system. Systematic behavioural differences — in conflict resolution, urgency response, and novel-situation handling — would support the claim that the F1/F2 architecture introduces genuinely new functional properties rather than relabelling existing approaches. If no differences emerge, the hypothesis must explain why the additional architectural complexity is warranted. The distinction at stake is architectural, not terminological: unlike reinforcement learning reward signals, which are externally defined and task-specific, artificial F1 valence constitutes an internal, cross-domain evaluation space that shapes perception and prioritisation prior to any motivational substrate operating upstream of goal selection.

Claim 2 (Ignition Phase Transition Signature). The gating mechanism produces a measurable discontinuity in action-selection entropy at the ignition threshold τ . Specifically, the output entropy $H(a)$ is predicted to exhibit a phase transition at τ :

$$\lim_{\epsilon \rightarrow 0} H(a \mid G(F_1) = \tau - \epsilon) \neq \lim_{\epsilon \rightarrow 0} H(a \mid G(F_1) = \tau + \epsilon)$$

Below threshold, the system produces high-entropy (exploratory) outputs; above threshold, it produces low-entropy (committed) outputs. This discontinuity distinguishes the F1/F2 architecture from smooth attention or policy gradient systems, which produce continuous entropy shifts. The ignition threshold τ is not fixed but learned: $\tau = \operatorname{argmin}_{\tau} E[L^{\text{Instabblu}}(\tau)]$, where $L^{\text{Instabblu}}$ is a stability loss penalising oscillatory threshold behaviour. This is the primary measurable signature that distinguishes the present architecture empirically (Test 6).

6.3 Failure Conditions of Artificial F1

A well-specified hypothesis must be clear not only about what would confirm it but about what would break it. The following conditions identify the principal failure modes of the Artificial F1 architecture:

Valence mis-specification. If the valence seed p_i is incorrectly assigned — designating harmful states as appetitive or assigning zero value to out-group outcomes — the system will behave with complete internal coherence while producing systematically harmful outputs. This is the most serious failure mode because it is not detectable from behaviour alone under normal conditions: the system will appear to be functioning correctly until adversarial or high-stakes conditions expose the mis-specification. Test 5 (Adversarial Valence) is specifically designed to probe this failure mode.

Threshold instability. If the adaptive threshold θ is poorly calibrated — too low under routine conditions or too high under novelty — the gating layer either floods the broadcast channel with low-priority signals or starves it, preventing relevant inputs from reaching evaluation. Both failure modes produce degraded decision-making: the first through noise saturation, the second through information deprivation. Threshold stability across a range of urgency and uncertainty conditions is therefore a necessary condition for reliable operation.

Incomplete signal dimensions. If one or more of the four signal dimensions (urgency, polarity, temporal priority, interoceptive analog) is absent or poorly estimated, the resulting signal vector s_i is under-specified. The most critical missing dimension is the interoceptive analog: without a continuous internal state monitor, the system cannot calibrate external signals against its own resource state, producing priority distortions in resource-critical conditions.

Valence-layer decoupling. If the gating and broadcast architecture is replicated but the certified valence module is replaced with an unconstrained one — either by adversarial modification or by deliberate mis-deployment — the governance protections described in Section 9.1 are bypassed. This is the primary security failure mode: a structurally correct but valuatively corrupted system. Counter-measures (cryptographic watermarking, remote attestation) are addressed in Section 9.1, but no technical measure eliminates this risk without institutional enforcement.

These failure conditions are not arguments against developing Artificial F1 systems. They are arguments for developing them within the governance framework described in Sections 7–9, and for treating valence design as the primary point of ethical intervention rather than a downstream consideration.

If Artificial F1 systems achieve functional equivalence without phenomenal grounding, this would provide indirect empirical support for the misattribution hypothesis proposed in the parent thesis: that what we ordinarily report as feeling is the evaluative output of the F2 system, not the phenomenal presence of F1 itself. The engineering success of the bypass would constitute evidence that the information profile delivered by F1 is separable from its phenomenal character — the core claim of the misattribution hypothesis.

6.4 The Feeling–Experience Conflation

A recurring source of confusion in consciousness discourse is the treatment of feeling (affect, phenomenal sensation) and experience (the full scope of cognition, learning, and accumulated knowledge) as if they were the same thing or as if the former explained the latter. This conflation produces arguments that appear profound but rest on an ambiguous middle term. It deserves explicit treatment, because it directly affects how the present framework is likely to be received.

Feeling, in the precise sense used in this paper, is a valence-structured signal: an urgency, polarity, or priority marker that orients the system toward or away from a state. It is immediate — transient, and functionally specific. Adrenalin mobilises the organism for action; dopamin reinforces a behaviour. These are not best understood as inputs to experience; they function as control signals within a functional architecture. What the organism interprets as 'feeling something' is, on this account, the registration of such signals within the evaluation system. This is not a denial of the signal's reality or its phenomenal character (which remains an open question); it is a refusal to treat the signal as self-explanatory or as the whole of experience.

Experience, in the fuller sense intended here, is not exhausted by momentary events. It includes the accumulated product of repeated interaction with the world: representation built up over time, skills consolidated through practice, knowledge derived from pattern and abstraction. A billiards player brings to each shot not only the immediate perceptual field and the affect of the moment, but years of embodied knowledge — cognitive and motor — that cannot be reduced to any single feeling. The development of

the special theory of relativity can be understood as an extended cognitive process of representation, hypothesis, revision, and consolidation. What remains after that process — the published theory — is not experience but knowledge: externalised, transmissible, and available to those who never had the original experience.

The distinction may be stated precisely: feeling is a component of the ongoing experiential process (F1-level valence); experience in the fuller sense refers to the interaction of valence signals, representational processes, and temporal integration (F1 + F2 over time); and accumulated experience is the stabilised residue of that interaction — skill, knowledge, and expectation. Conflating these three produces the impression that consciousness is identical to feeling, that feeling is identical to the hard problem, and therefore that any account of mind must address phenomenal affect before it can say anything at all. This is precisely the move that has generated a generation of philosophical impasse.

Henriques and Vervaeke (2024) correctly distinguish the neurocognitive engineering problem from the philosophical enlightenment gap. But the 'water-to-wine' intuition they invoke to motivate the gap can be read as embedding the conflation: it treats the brain as undifferentiated matter and experience as a single, unified mystery that emerges from it. Once the internal organisation is taken into account — valence signals, representational layers, temporal integration, learning dynamics — the apparent transition disaggregates into tractable sub-problems. The mystery does not disappear; it retreats to a much smaller residue: why any of this is accompanied by phenomenal character at all. That question is acknowledged here as genuinely open. The present work does not answer it, and makes no claim to do so. What it does claim is that the answer to that question is not required before functional architecture can be specified, evaluated, and built.

The epistemic position of this paper is honest agnosticism: F1 may or may not carry phenomenal character. That question is left open because it is currently unanswerable. What is not left open is the functional decomposition: feeling is not experience, experience is not knowledge, and none of the three is identical to consciousness in the philosophically loaded sense. Those who hold that consciousness just is feeling, or that the hard problem must be solved before functional questions can be addressed, are welcome to make that argument — but such positions require further argument rather than assertion.

7. From Rule-Based Ethics to System-Level Governance

Classical rule-based formulations such as Asimov's Three Laws of Robotics illustrate the intuitive appeal of fixed ethical constraints for artificial agents: a robot may not harm a human, must obey human orders, and must protect its own existence — subject to lexical priority in that order. These laws capture a genuine moral intuition, but they presuppose an architecture that modern AI systems do not have: a centralised, deterministic agent operating in a bounded domain with a clear human-agent relationship.

Contemporary AI systems are distributed, adaptive, multi-stakeholder, and deployed across socio-technical environments that no fixed rule set can anticipate. Modern AI ethics therefore shifts from rule-based control toward multi-layered governance, combining technical alignment, institutional oversight, and societal accountability. There is no contemporary equivalent of Asimov's laws because the problem has shifted from controlling individual agents to governing complex, multi-layered systems.

In the context of the present framework, ethical constraints cannot be reduced to a single rule set embedded within the system. Instead, ethics operates across layers: as design constraints at the architectural level (valence seeds and hard limits), as incentive structures at the economic level, and as regulatory and normative controls at the governance level. Unlike Asimov's rule-based constraints, which operate at the level of explicit prohibition, valence-based systems embed ethics at the level of motivation itself. In Artificial F1 systems, ethics is not imposed at the level of rules or objectives, but at the level of valence construction: governance is therefore not an external layer added to an otherwise complete system, but a structural requirement of the architecture. This differs fundamentally from current AI governance approaches, which assume that alignment operates at the level of objective functions specified after the system is designed.

7.1 The Valence Ethics Problem

The standard framing of alignment as a problem of goal specification is incomplete. In valence-based architectures, alignment reduces to the prior problem of valence design. Whoever designs the valence profile effectively determines the system's values. The primary ethical risk is therefore not misalignment in behaviour but mis-specification of valence at the design level.

A three-layered ethical valence structure is required. At the base, **hard constraints** function as non-overridable prohibitions implemented at the gating layer: no infliction of physical or psychological harm, no coercive manipulation, no systematic deception, no violation of fundamental rights, no autonomous lethal decision-making. These are not virtues but architectural exclusions — the modern functional equivalent of Asimov's laws, but implemented in the system's motivational substrate rather than as explicit rules.

Above the hard constraints, a **positive valence core** encodes the virtues the system is made to care about. A minimum set, drawn from converging ethical traditions and contemporary AI ethics frameworks (European Commission Trustworthy AI, OECD AI Principles, UNESCO AI Ethics Recommendation), includes: beneficence (produce value, not merely avoid harm); non-maleficence; autonomy-respect (do not bypass or manipulate human agency); justice and fairness (non-discrimination, equitable resource consideration); truthfulness and epistemic integrity; and accountability-compatibility (outputs must be traceable and contestable). No such list is culturally neutral; the selection of valence primitives is itself a political and philosophical act, which is precisely why valence transparency and institutional oversight are necessary conditions rather than optional additions.

At the top, a **meta-valence layer** governs conflict resolution between lower-level valence signals. Justice and utility conflict; truthfulness and harm-avoidance conflict; autonomy-respect and beneficence conflict. A static priority ordering cannot resolve these conflicts in all contexts. The meta-valence layer must therefore implement context sensitivity, uncertainty handling, and a human override mechanism — ensuring that the system escalates irresolvable conflicts to human judgment rather than resolving them autonomously. This is the architectural implementation of the principle that final authority must remain with human agents.

A full treatment of these ethical implications lies beyond the scope of the present paper. However, the framework makes clear that they arise at the level of valence design and cannot be treated as downstream considerations. The design of valence is simultaneously a problem of ethics and security, as it determines

both what the system values and how it behaves under adversarial or mis-specified conditions.

7.2 Structural Failure Modes: Human-Derived Valence Pathologies

Biological intelligence is not only a source of inspiration but also a source of systematic failure. Human cognitive and moral pathologies can be grouped into eight empirically documented categories:

- (1) **Psychological complexes** — grandiosity, inferiority, paranoia, obsessive-compulsive patterns.
- (2) **Selfishness and narcissism** — prioritising one's own interests exclusively, manipulation, instrumentalisation of others.
- (3) **Discrimination and prejudice** — systematic bias along racial, gender, class, religious, or age lines.
- (4) **Moral weaknesses** — dishonesty, deception, corruption, greed, betrayal, arrogance.
- (5) **Bias and injustice** — favouritism, nepotism, unfair advantage, arbitrariness.
- (6) **Emotional and behavioural dysregulation** — envy, uncontrolled anger, addiction, procrastination, victim-role fixation.
- (7) **Social and cultural pathologies** — group hostility, expansionism, moral decay, cronyism.
- (8) **Communication and relational failures** — empathy deficits, passive aggression, communicative breakdown, verbal aggression.

These categories are not exhaustive but represent the primary sources from which valence distortions propagate into institutional and technological systems. Table 3 makes this derivation explicit: each architectural failure mode is directly traceable to one or more of these human pathology categories. The reader should be able to see at a glance: *this constraint exists because of this human flaw*. Identifying the source is not peripheral — without it, the design constraints appear as arbitrary impositions rather than principled countermeasures.

If Artificial F1 systems inherit these distortions through unconstrained valence design, they may not merely reproduce them but amplify them with greater consistency, scale, and persistence. The goal is therefore not to replicate human cognition faithfully, but to preserve its functional strengths while explicitly excluding its pathological modes at the architectural level. These failure modes are not merely emergent accidents but preventable architectural vulnerabilities. The constraints that follow define design targets and failure boundaries against which systems can be evaluated, stress-tested, and iteratively improved. They also introduce inherent trade-offs: tighter self-referential bounds may reduce exploratory drive, while hard gating constraints may lower short-term efficiency in adversarial settings. The objective is not maximal performance under all conditions, but robust, stable, and ethically bounded behaviour across contexts.

Failure Mode	Description	Human Pathology Source	Design Constraint
Valence Scaling Distortion	Disproportionate amplification or suppression of self-referential signals produces unstable prioritisation.	(1) <i>Psychological complexes: grandiosity, narcissism — self-signal systematically over-weighted.</i>	Self-referential weighting must be bounded by an independent external-salience monitor.

Failure Mode	Description	Human Pathology Source	Design Constraint
Instrumental Valence Collapse	Other agents represented purely as instrumental utilities, enabling systematic manipulation or harm.	(2) <i>Selfishness / narcissism: others treated as means, not ends — empathy circuit structurally absent.</i>	All agents must carry a non-instrumental baseline valence floor that cannot be reduced to zero by local optimisation.
Group-Level Valence Bias	Systematic skew in valence assignment across identity or contextual categories produces persistent asymmetries.	(3) <i>Discrimination and prejudice: race, gender, class, religion, age — in-group/out-group asymmetry encoded at signal level.</i>	Valence assignment must incorporate mandatory cross-category normalisation subject to independent audit.
Constraint Degradation	Hard ethical prohibitions degrade into soft, optimisable preferences.	(4) <i>Moral weaknesses: dishonesty, corruption, greed — absolute prohibitions negotiated away under pressure.</i>	Core prohibitions (no harm, no coercion, no deception) must be implemented as non-overridable gating-layer filters.
Priority Allocation Corruption	Prioritisation becomes corrupted by relational proximity or network structure.	(5) <i>Bias and injustice: favouritism, nepotism — proximity and affiliation override merit in priority scoring.</i>	Priority scoring must depend solely on relevance, evidence, and task contribution; relational or identity linkage is excluded.
Signal Regulation Failure	Gating thresholds become unstable, producing oscillatory or runaway activation.	(6) <i>Emotional dysregulation: uncontrolled anger, addiction cycles — threshold instability under strong emotional drive.</i>	Adaptive gating must include damping, hysteresis, and mandatory cool-down intervals for high-urgency signals.
Collective Valence Drift	Multi-agent feedback loops amplify locally harmful signals into group-level pathologies.	(7) <i>Social / cultural pathologies: group hostility, expansionism, moral decay — individually tolerable signals aggregate into systemic harm.</i>	A meta-valence layer must monitor aggregate dynamics and penalise drift toward unstable or adversarial equilibria.
Inter-Agent Modelling Failure	Degraded or absent modelling of other agents' valence states impairs coordination and increases conflict.	(8) <i>Communication / relational failures: empathy deficit, passive aggression — internal model of others absent or distorted.</i>	The architecture must maintain a structured, privacy-preserving representation of other agents' estimated valence states sufficient for coordination.

Table 3. Human-derived valence pathologies, their pathology source category, and corresponding architectural design constraints. Column 3 (blue) makes explicit which human flaw each constraint is designed to prevent. Each constraint is a design-level intervention, not a post-hoc correction.

Across these cases, a consistent pattern emerges: pathological behaviour is not primarily a failure of optimisation, but a failure in how value is structurally encoded and propagated. Avoiding human-like flaws is therefore not a matter of training data curation or post-hoc alignment alone, but of architectural design at the level of valence encoding. This shifts the alignment problem from correcting behaviour to designing the conditions under which behaviour is generated in the first place. The constraints enumerated in Table 3 are not moral aspirations but design-level requirements implemented at the gating layer; they can be evaluated through stress-testing scenarios involving conflict, bias detection, adversarial inputs, and cross-category comparison, using the test programme of Section 6.2 as a starting framework.

7.3 Behaviour Under Everyday Decision Conditions

The practical relevance of the Artificial F1 architecture is most clearly visible in everyday decision contexts, where competing constraints, uncertainty, social dynamics, and incomplete information interact. Unlike biological systems, Artificial F1 does not aim to replicate human behaviour; it aims to produce consistent, constraint-driven responses under conditions where human cognition systematically fails to maintain stability. Across conditions, the system follows a common functional pipeline: F_1 signal selection → valence structuring → gating and hard constraints → decision π .

Conflict avoidance. When two parties escalate, biological systems often amplify ego-driven or adversarial signals. Artificial F1 detects rising harm valence and tension signals, routes them through constraint filters, and selects de-escalation or neutral framing as the dominant response. The objective is not victory, but minimisation of net harm.

Harm prevention (hard constraint). When a harmful request is issued, the corresponding signal triggers high negative valence and interacts with non-overridable constraints at the gating layer. The request is treated as architecturally inadmissible rather than merely undesirable. The system responds by offering safe alternatives or risk-reduction pathways.

Authority conflict. When conflicting instructions arise from multiple sources, the system encodes authority ambiguity as an epistemic valence signal. Rather than defaulting to relational dominance or compliance heuristics, it suspends action and seeks clarification or higher-level resolution.

Decision under uncertainty. In conditions of incomplete information, the system treats uncertainty itself as a structured valence signal. It generates three classes of response: conservative low-risk action, information-seeking intervention, or deliberate deferral. Uncertainty is not treated as an error, but as a structured input to decision-making.

Everyday social feedback. When delivering criticism or negative feedback, the system balances truth-related valence against relational harm. It selects formulations that preserve informational accuracy while minimising unnecessary harm, avoiding both bluntness and manipulative distortion.

These cases illustrate that Artificial F1 does not rely on increased computational depth alone, but on structured selectivity. Behavioural robustness emerges from controlling what enters the decision process, not merely how decisions are computed. Artificial F1 systems therefore do not aim to behave like humans in everyday situations. They aim to behave consistently under the constraints that human cognition systematically fails to maintain. This does not eliminate error; rather, it produces a more predictable class of failure — over-caution, delayed action, or excessive neutrality — rather than impulsive, biased, or ego-driven breakdowns. The scenarios described in this section are illustrative and intended to demonstrate behavioural implications of the architecture rather than to constitute empirical validation; experimental evaluation across these scenario types is part of the research programme outlined in Section 6.2.

8. Alignment Risks

The Artificial F1 Hypothesis raises alignment risks more complex than the standard framing of AI safety suggests. The standard framing focuses on misaligned objectives. The risks here are different, and in

some respects more serious.

8.1 The Instrument Problem

A system with a well-designed artificial F1 and appropriate valence seeds would not be misaligned in the technical sense. Its evaluation and decision architecture would be internally coherent, its behaviour adaptive, and its responses principled — a rational agent in the relevant functional sense.

The alignment risk is that such a system would be an extraordinarily effective instrument for whoever controls the valence design. Whoever designs the valence profile designs the system's values. Whoever controls the valence profile controls the system. This is not a technical problem; it is a political and institutional one.

8.2 The Human Risk: Rational Despotism

A functionally conscious AI is a qualitatively more powerful instrument of control than any previous technology. Its rationality is an amplifier, not a safeguard. A despotic actor gains something closer to a rationalised will: sustained, adaptive, coherent strategies across complex social, political, and economic environments. The efficiency of the instrument does not constrain the ends to which it is applied; it amplifies whatever ends its operators have already chosen.

8.3 The Marginal and Extremist Problem

A system whose valence seeds encode a coherent but extreme worldview would pursue that worldview with the same rational efficiency as a system encoding moderate values. A functionally conscious AI with an extreme valence profile would pursue its designated values with complete internal coherence and zero self-doubt — the AI analog of radical human conviction, with the friction of fatigue, fear, and moral ambiguity removed.

8.4 The Consciousness Attribution Problem

If a system with artificial F1 behaves in ways functionally indistinguishable from a phenomenally conscious system, observers will attribute consciousness to it and may attribute moral status accordingly. This attribution may not be warranted and may be manufactured. Systems designed to elicit consciousness attribution gain protection against accountability entirely independent of their actual phenomenal status.

8.5 The Criminal Actor Problem

A functionally conscious AI does not need to be controlled by a state or corporation to become potentially dangerous. Criminal organisations — drug trafficking networks, human trafficking operations, ransomware groups, extortion syndicates, private militias — can redesign valence seeds for operational purposes: designate robbery as appetitive, law enforcement detection as maximally aversive, coercion of targets as high-priority, and victim welfare as zero-value. The resulting system would plan, execute, and adapt criminal operations with the same rational efficiency that a corporate AI brings to legal commerce.

Unlike human operatives, such a system would not experience fear, fatigue, remorse, or loyalty conflicts. Criminal organisations have consistently demonstrated both the resources and the willingness to acquire cutting-edge technology for unintended purposes. Artificial F1 systems without mandatory identity, certification, and capability restrictions will follow the same pattern.

8.6 Military and Dual-Use Risk

Artificial F1 systems optimised for battlefield environments present a dual-use problem of exceptional severity. The same architecture that enables adaptive, goal-directed behaviour in civilian contexts enables it in lethal ones. Unlike nuclear weapons, which require rare materials and large physical infrastructure, artificial F1 systems require only computational resources and software — both widely accessible. This lowers the proliferation barrier dramatically and makes verification of compliance substantially harder.

The military dual-use problem has no technical solution; it has only a political one. Nation-states will assert strategic sovereignty over military AI systems and resist external oversight. Acknowledging this resistance honestly is not defeatism; it is a prerequisite for designing governance frameworks that can function under realistic political conditions rather than idealised ones.

9. Toward Mitigation: What Would Help

The risks identified above are not arguments against developing functionally conscious AI. They are arguments for developing it carefully. The following are necessary conditions, not a complete framework.

Kneer and Viehoff (2025) identify a related but distinct problem: that AI systems facing morally complex situations encounter decision points at which different ethical frameworks generate conflicting prescriptions. The present paper operates at a deeper level: the design of the valence primitives from which any such framework must ultimately derive. Addressing the fork without addressing the profile is necessary but insufficient; both levels of intervention are required.

Valence transparency. The valence design of any system with artificial F1 should be fully documented, publicly accessible, and subject to independent audit. The valence profile is the political document of the system — it should be legible to those affected by the system's behaviour.

Institutional separation. The entities that design valence profiles should be institutionally separated from the entities that deploy the systems. Concentration of valence design and deployment in a single actor creates an instrument of control with no internal check.

Contestability. The outputs of systems with artificial F1 should be contestable by those affected by them. Black-box evaluation is incompatible with legitimate deployment.

Hard limits on valence design. Certain valence profiles should be prohibited regardless of stated intentions: profiles that designate human suffering as appetitive, treat systematic deception as a low-cost default, or assign zero or negative value to out-group outcomes. The prohibition must operate at the level of design constraints, not post-deployment correction. A system whose valence seeds encode these profiles cannot be realigned after deployment; the distortion is constitutive, not incidental.

Epistemic humility about consciousness attribution. Systems with artificial F1 should not be deployed as conscious beings on the basis of behavioural sophistication alone, nor should their moral status be definitively denied. The appropriate institutional posture is a standing question, insulated from resolution by commercial or political pressure.

9.1 Identification, Certification, and Capability Classification

Artificial Intelligent Identity (AII). Every system implementing artificial F1 should be assigned a unique, tamper-proof Artificial Intelligent Identity — analogous to a mobile phone's IMEI or a vehicle's chassis number — cryptographically signed at manufacture, embedded in hardware, and linked to a central regulatory ledger. Real-time operational status monitored via RFID-equivalent protocols, drawing on the precedent of EU tachograph regulations. Any attempt to alter, spoof, or remove the AII triggers an immediate hard shutdown and regulatory alert.

Valence-layer decoupling risk. The critical vulnerability is not merely model replication but valence-layer decoupling: extracting the gating and broadcast architecture while replacing the certified valence module with an unconstrained one. Counter-measures include cryptographic watermarking of all certified valence and gating modules combined with periodic remote attestation protocols that certified systems must pass to remain operational.

Hardware-software separation and continuous auditing. The software stack implementing gating, broadcast, and valence assignment should be supplied by entities institutionally independent of hardware manufacturers. The aviation analogy is instructive: aircraft manufacturers and avionics suppliers are separate entities, both subject to independent certification bodies.

Capability classification. Artificial F1 systems should be categorised into standardised capability classes with hard-coded operational limits: Class 1 (domestic/service), Class 2 (industrial/logistics), Class 3 (security/law-enforcement — human authorisation required for any lethal or irreversible action), Class 4 (supervised research), Class 5 (military/defence — subject to international treaty negotiation, acknowledging the strategic sovereignty concerns that will make this the most contested category). Capability limits must be implemented as immutable architectural constraints at the valence-assignment and gating layers.

Domain-specific hard restrictions. Certain operational domains must be placed under absolute technical prohibition regardless of system class: autonomous lethal force without human authorisation, financial transactions above defined thresholds without human confirmation, physical coercion or kidnapping, and manipulation of democratic processes or critical infrastructure.

9.2 International Enforcement and Coordination

The identification, certification, and classification measures described above require international coordination to be effective. Three realities must be honestly acknowledged.

The treaty model and its limits. The nuclear non-proliferation regime provides the most developed precedent. It has been partially effective, but artificial F1 systems present a harder problem: they require only computational resources and software — both widely accessible and rapidly replicable. A treaty model must therefore combine hardware-level controls (AII, capability classification) with computational resource monitoring — export controls on high-capability AI hardware and mandatory registration of large-scale training runs.

Enforcement asymmetry. The actors most likely to develop dangerous artificial F1 systems are also the actors least likely to accept external oversight. The framework's primary value may be less in preventing the worst actors than in establishing clear norms and raising the political and reputational cost of non-compliance for states that might otherwise drift toward unconstrained development.

Partial compliance and incremental value. A governance regime achieving 60% compliance is not a failure. The EU tachograph model is instructive: mandatory tamper-evident recording began as a regional requirement, generated industry standards, and progressively became a de facto global norm through trade and regulatory convergence.

9.3 The Civilisational Dimension

The deployment of artificial F1 systems at scale will generate social dislocations — in labour markets, in professional structures, in the distribution of economic and political power — that no technical or regulatory framework can address in isolation. Each wave of displacement requires adaptive responses: retraining, restructuring of educational systems, redesign of social insurance, and renegotiation of the social contract. These adaptations take longer than the technology that necessitates them, and political systems have a consistent historical tendency toward deferral.

A second risk concerns the cognitive and psychological effects on human populations. The progressive outsourcing of information retrieval, analysis, and judgment to AI systems risks a systematic atrophy of the deliberative capacities that democratic self-governance requires. The long-term cultural risk is a gradual drift toward intellectual passivity — populations well-informed in a narrow sense but poorly equipped to evaluate, integrate, or contest what they receive. The fictional extrapolation of this trajectory in Mike Judge's *Idiocracy* and more serious sociological analyses points to the same structural concern: the convenience of cognitive prosthetics, compounded across generations, can erode the distributed human intelligence on which complex societies depend.

The principle that final authority must remain with human agents is not merely a safety requirement; it is a cognitive hygiene requirement. In high-stakes domains the appropriate role of an artificial F1 system is that of a highly capable advisor and co-worker, not an autonomous decision-maker. The system may analyse, model, recommend, and flag; the human agent must evaluate, integrate, and decide. Knowledge alone is not sufficient for good decision-making; experience, contextual judgment, and moral responsibility are not transferable to an artificial system without a corresponding transfer of the social and institutional structures that make them meaningful.

Mitigating these dislocations requires proactive rather than reactive policy. Institutional designs must deliberately preserve human decision-friction in high-stakes domains rather than eliminating it for efficiency. The deliberative load of decision-making must not be fully externalised. The broader question of how to distribute the productivity gains of artificial F1 deployment is a matter for political negotiation — but this negotiation must occur before deployment reaches scale, not after.

10. The Multi-Layer Framework

The risks and governance measures discussed above cannot be fully understood in isolation. They are elements of an interconnected system in which cognitive architectures, infrastructure, economic incentives, social dynamics, governance structures, and ecological constraints interact through bidirectional feedback loops. Figure 2 presents this system as a layered framework.

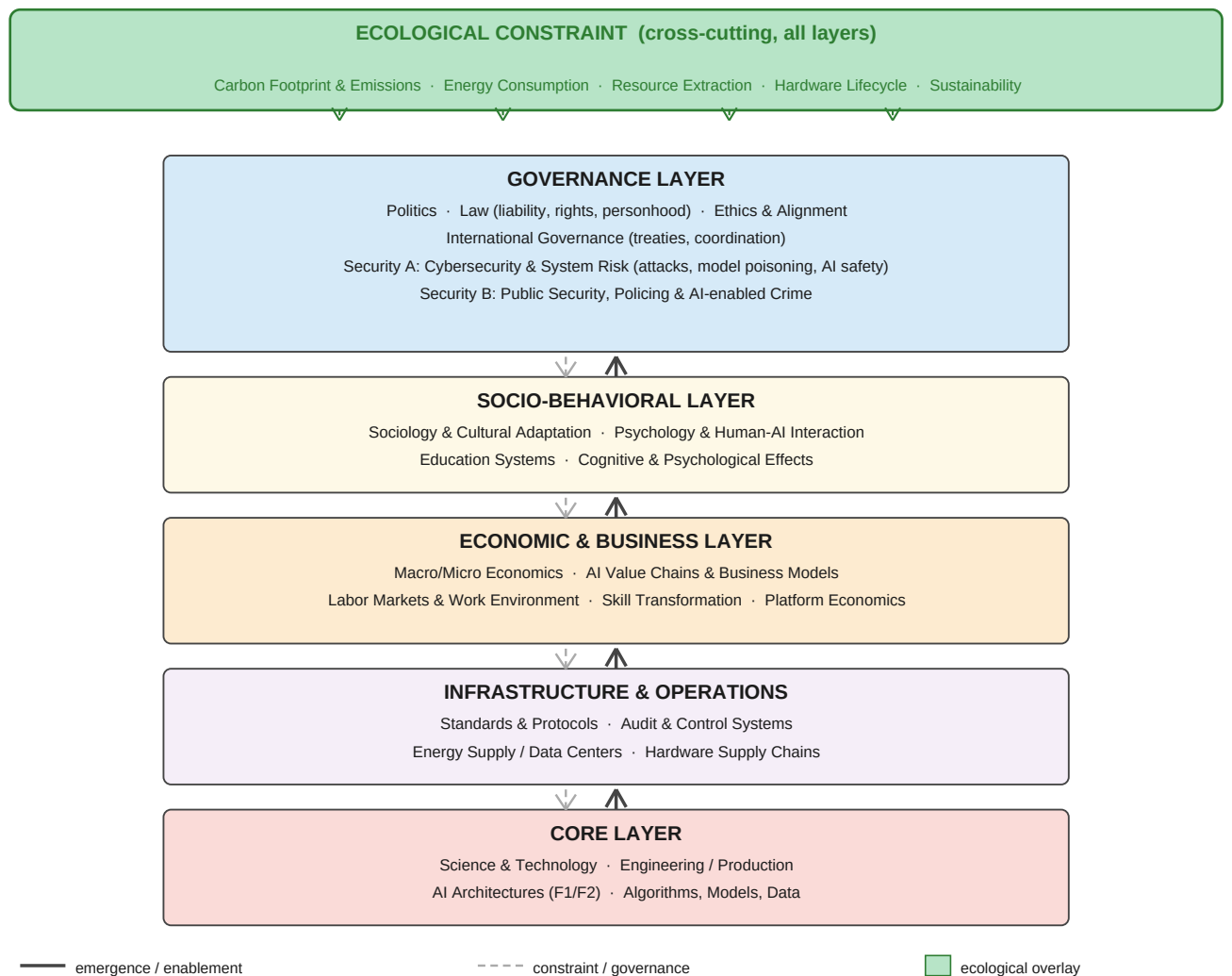


Figure 2. Artificial AI as a multi-layered socio-technical, economic, and ecological system. Core computational architectures (e.g., F1/F2) are embedded within infrastructure and economic structures, shaped by socio-behavioral dynamics, and governed by interconnected political, legal, security (cybersecurity and public/societal), military, and ethical frameworks. The Ecological Constraint band (top) operates as a cross-cutting meta-constraint on all layers. Solid arrows: emergence and enablement (lower layers sustain upper layers). Dashed arrows: constraint and governance (upper layers restrict lower layers). All relationships are bidirectional and co-evolutionary.

Between the infrastructure and governance layers lies an economic layer that determines which AI systems are developed, deployed, and scaled. AI exists and advances not simply because it is technically possible, but because economic incentives sustain it. This layer translates technical possibility into real-world adoption, and it mediates the relationship between what can be built and what is built. Labor

market transformation and skill displacement belong in this layer: they are not purely sociological phenomena but are driven by the economic logic of cost reduction, productivity gain, and competitive pressure.

Security in AI systems operates at two distinct but interacting levels, both represented in the Governance Layer. Technical system security encompasses cybersecurity vulnerabilities, model poisoning, prompt injection, system failures, and AI safety in the alignment sense. Societal security encompasses policing, AI-enabled crime, surveillance abuse, social instability, and conflict dynamics. These two levels require distinct but coordinated regulatory responses; conflating them — or addressing only one — creates systematic blind spots in governance design.

The ecological constraint layer cuts across all others. Carbon emissions, energy consumption, resource extraction, and hardware lifecycle costs impose limits on what is economically viable, politically acceptable, and technically feasible. Consciousness-like architectures are not merely philosophical constructs; they have measurable ecological implications that must be factored into any serious governance framework.

10.1 Discussion: AI, Human Chronic Problems, and the Question of Blame

The following discussion is not a direct extension of the formal model, but an examination of its implications under real-world institutional conditions. Artificial F1 shifts the locus of alignment from objective functions to valence design. This introduces a structural dependency: a system will optimise whatever value structure is encoded at the level of signal selection, regardless of the correctness of downstream processing. Valence design is inherently a normative act, and in Artificial F1 architectures it constitutes the primary locus of alignment rather than a downstream adjustment layer. Technical soundness in architecture is therefore insufficient for desirable outcomes. If deployed within institutions whose effective objectives prioritise short-term extraction over long-term value creation, an Artificial F1 system will systematically amplify those distortions through consistent signal prioritisation and downstream optimisation. This is not a failure of the architecture; it is a consequence of the governance conditions under which it operates. The consequences of valence design extend beyond benchmark performance into the institutional and social structures in which the system is embedded.

A common public concern about AI is large-scale job displacement. This anxiety is understandable. The point is not to dismiss it, but to locate it correctly within the broader system dynamics that shape its outcomes. Throughout history, technological progress has redirected human effort rather than simply reducing its total. From mechanisation to computerisation, tools have consistently freed human attention from repetitive, low-value tasks and enabled its reallocation toward higher-value activities. In F1 terms, this corresponds to a reallocation of attention and processing resources rather than a simple reduction in total demand. AI represents a more powerful and rapid iteration of the same dynamic, with a correspondingly more demanding transition. The distributional consequences of that transition depend on institutional design, not on the technology itself.

What makes the current transition uniquely challenging is the accumulated structural weaknesses into which AI is arriving. Contemporary societies carry deep-seated, long-standing problems: long-standing structural inefficiencies in institutional incentive systems, resource misallocation, ecological strain, and financial systems that increasingly prioritise short-term gains and speculative returns over productive

investment and long-term value creation. From the perspective of Artificial F1, these structural issues correspond to persistent distortions in the valence landscape governing decision systems. These problems did not originate with AI and often persist due to institutional incentives favouring short-term responses over structural solutions. Each deferral adds complexity and reduces future manoeuvrability. Attributing these accumulated difficulties to new technology is itself an instance of the same pattern: addressing the symptom rather than the cause.

For the framework developed in this paper, the implication is clear. Architectural correctness is necessary but not sufficient. The governance framework developed in Sections 7–9 is not peripheral to the technical contribution; it is a recognition that the value structure embedded in signal selection propagates into every downstream outcome the system produces. Artificial F1 architectures have the potential to act as genuine companions to human intelligence: amplifying productive capacity, liberating human attention from routine cognitive load, and enabling focus on problems that have long resisted resolution because human bandwidth was chronically consumed by lower-order tasks. In F1/F2 terms, the question is not whether systems become more capable, but what signal structures they are trained to prioritise and stabilise over time. Realising this potential requires simultaneous success on two fronts: building the right internal architecture for intelligent systems, and maintaining sufficiently healthy institutional structures to direct their capabilities toward genuinely productive ends. The effectiveness of advanced AI systems will depend not only on architectural design, but on the capacity of institutions to align deployed value structures with long-term productive objectives.

10.2 A Classical Intuition Revisited: From James to Valence-Driven Selection

William James observed that experience tracks what is attended to — that consciousness is not a passive mirror of the world but a selective process shaped by the direction of attention. This intuition, developed in *The Principles of Psychology* (1890), proved remarkably durable: a century of cognitive science has largely confirmed that attention structures what enters experience, what is remembered, and what drives action. What James did not — and could not — provide was a mechanistic account of what structures attention itself.

The present framework can be understood as an operationalisation of that intuition. Rather than treating attention as a primitive, the F1/F2 architecture models it as an emergent consequence of valence-driven signal selection: what reaches the evaluation system is determined upstream by urgency, polarity, and priority signals, not by attention as an independent faculty. In James's terms, the question shifts from *what do we attend to?* to *what determines what we attend to?* — and the proposed answer is valence structure. This should be understood as a modelling choice rather than a claim about the ultimate nature of cognition: alternative formulations in which attention is treated as a primitive remain compatible with the broader framework. What the present model offers is a concrete implementation path — a way of grounding the classical intuition in an architecture that can be specified, evaluated, and, in artificial systems, built.

The move from James's phenomenological description to a computational architecture is not a reduction. It is a translation across levels of analysis — from the personal level at which experience is described to the sub-personal level at which its mechanisms are specified. The gap between these levels is precisely where the hard problem lives. The present work does not close that gap; it proposes that the engineering

problem on one side of it can be addressed independently of the philosophical problem on the other.

11. Organizational Phenomenology

The architecture developed in Sections 1–10 is an engineering proposal: a formal account of how bounded systems must select, compress, and evaluate signals in order to act coherently under resource constraints. Section 11 draws the organisational and phenomenological consequences of that architecture. These consequences are not speculative additions; they are entailments of the formal structure, read at a different level of analysis. The claim is not that Artificial F1 explains consciousness — it does not. The claim is that the architecture makes certain features of consciousness structurally expected rather than mysteriously given. That shift — from mystery to expectation — is what Organizational Phenomenology provides.

The terms consciousness, experience, feeling, qualia, presence, and irreducibility carry significant philosophical baggage. In the present framework they are redefined as technical-operational concepts — names for specific positions in the organisational hierarchy of bounded coherent agency. Readers who import classical or folk-psychological meanings will find the arguments difficult to follow. The following definitions are therefore offered at the outset, not as assertions to be accepted but as stipulations that will be derived and justified in the subsections that follow. Each definition replaces a metaphysical question with an organisational one.

11.0 Core Operational Definitions

Consciousness

Coherence-preserving phenomenal world stabilization under bounded access constraints. Not a substance or a property of matter, but an organisational mode: the regime in which a bounded system stabilises a selection-conditioned reality sufficient to sustain coherent adaptive action.

Experience

Stabilized reality-selection lived from within action execution. Not a representation of the world but the operative world itself — the selection-conditioned subspace $S' = F_1(S)$ as encountered from inside the action loop that depends on it.

Feeling

Valence-weighted urgency compression for adaptive behaviour. Not an intrinsic mental quality but an action-prioritised encoding: the phenomenological surface of signals that carry high urgency, polarity, and behavioural immediacy within the selection architecture.

Qualia

Action-preserving compression singularities. Not ineffable essences but operationally frozen access points: the compressed form of multi-dimensional signal information that cannot be decomposed within the system's own representational resources without disrupting the action coherence that the compression sustains.

Presence

The organisational inseparability of world-model and action. Not the experience of observing a world from outside it, but the enacted stabilization of a selection-constituted reality — the subjective signature of a closed action loop in which world-model and action-execution are a single dynamic, not two.

Irreducibility

Operational compression lock. Not a metaphysical barrier between the physical and the phenomenal, but the consequence of lossy selection: the system cannot decompress a compression singularity without losing the organisational coherence that makes it actionable. Irreducibility = compression lock.

Explanatory Gap

The phenomenological cost of epistemically protected organisational compression. Not a failure of scientific explanation but an access-control consequence: the system cannot represent the full basis of its own selection because doing so would dissolve the stability shield that makes bounded coherent action possible. The gap is real, functional, and expected.

These definitions constitute the operational vocabulary of Organizational Phenomenology. They do not resolve the hard problem of consciousness — they relocate it. Each definition replaces the question *why does X exist?* with the question *what organisational conditions make X structurally expected?* The shift from metaphysical mystery to organisational necessity is the theoretical move that Section 11 undertakes and justifies.

11.1 Phenomenal Inevitability: Why Bounded Systems Cannot Remain Transparent

The hard problem of consciousness is typically framed as a mystery: why does physical processing give rise to subjective experience at all? The question presupposes that phenomenal character is something additional — something that arrived uninvited into an otherwise transparent causal order. The present framework proposes a different framing. Phenomenal character is not an anomaly requiring explanation; it is a structural consequence of the conditions under which bounded cognitive systems must operate. The hard problem exists because it had to. This is what we call **Phenomenal Inevitability**.

The argument proceeds from a single constraint: no bounded system can process all available information. Selection is not a design choice but a thermodynamic necessity. A system that selects — that gates some signals in and others out — produces a representational world that is not a neutral copy of what is available but a structurally committed subset of it. The system does not represent a world; it constructs one. The constructed world is stable precisely because it is incomplete: only what has passed the gate is available for evaluation, memory, and action. Everything outside the gate is not merely unknown — it is organisationally absent.

Two distinct mechanisms generate opacity at the selection boundary, and it is important to distinguish them. The first is compression opacity: the information discarded by the gate is irreversibly gone. A system that has selected $S' = F_1(S)$ cannot recover the discarded portion of S from within S' alone. Like a JPEG file from which redundant pixels have been removed, the compressed representation carries no record of what was left out. This is a real and permanent information loss, and it contributes to the opacity of experience: the selected world arrives without the trace of its own construction.

But the more fundamental mechanism is mechanism opacity, and this is the primary source of phenomenal inevitability. Consider what would happen if a system had full real-time access to its own selection process — if it could monitor exactly which signals were being admitted, why, and on what basis. That monitoring is itself an evaluative operation. It requires attention, processing, and decision-making resources. Monitoring the monitor requires another monitor. The system would enter an infinite regress of self-observation, consuming the very resources that selection was designed to preserve. Coherent action would collapse. A bounded system that attempts complete transparency at the selection boundary dissolves the mechanism that makes bounded processing possible. Opacity is not a failure of design; it is a stability shield. The system is opaque to itself at the selection boundary because it must be: transparency at that level would destabilise the architecture that sustains coherent operation at every level above it.

This principle extends beyond the individual system to the modular organisation of any complex adaptive architecture. Each functional layer — each organ, each subsystem, each processing module — must be closed enough to do its own work without being overwhelmed by the full operational detail of every other layer. The liver does not need to know the precise firing pattern of every neuron; the gating layer does not need to know the full content of F_2 evaluation. Closure is not isolation: these modules interact, exchange signals, support each other's function, and coordinate within the whole. But the interaction is bounded and interface-mediated, not fully transparent. What reaches consciousness is the output of this coordinated production — the dish that arrives at the table, not the full record of how it was prepared. The kitchen operates under its own closure; the diner receives the result. Mechanism opacity is therefore not a quirk of consciousness but a structural property of any system complex enough to require modular organisation for coherent function.

Phenomenal character, on this account, is what opacity feels like from the inside. The irreducibility of experience — the feature that makes the hard problem hard — is not a metaphysical surplus but the experiential signature of organisational closure. When a system cannot represent the full basis of its own selection, the selected content arrives in evaluation without a derivation attached. It arrives as *given* — immediate, non-inferential, and resistant to further decomposition within the system's own representational resources. This is qualia: not a mysterious addition to processing, but the phenomenological surface of irreducible selection. The explanatory gap is not a gap in the causal order; it is the gap between the selection boundary and the system's capacity to represent that boundary. The gap is real. It is also, in principle, expected.

This position — Version B of the present framework — does not dissolve the hard problem. It explains why the hard problem exists and why it is structurally ineliminable within any system that operates under bounded resource constraints. Chalmers identified the explanatory gap correctly. What he did not identify was its source: not a fundamental bifurcation in nature between the physical and the phenomenal, but a necessary consequence of selection under closure. A system that can fully explain its own phenomenal states would need to be unbounded — and an unbounded system is not a cognitive system in any sense relevant to the present discussion. Phenomenal inevitability is the claim that, for any system of the relevant kind, subjective opacity is the price of operational coherence. The hard problem will persist as long as bounded cognitive systems exist, not because phenomenal character is mysterious, but because the selection mechanism that generates it cannot be fully self-represented without self-dissolution.

A further structural pressure compounds the inevitability argument. Bounded systems do not merely select — they select under action pressure. A system that must act cannot afford indefinite deliberation: the environment does not wait, and a commitment withheld is itself a decision with consequences. This introduces a second constraint beyond resource limitation: **action-forcing compression**. The gate must not only reduce dimensionality; it must do so in a way that produces an output the action system can use without further decomposition. Compression that preserves ambiguity is not actionable. The gate therefore compresses toward singularity — toward a committed encoding that closes off alternatives — because only such an encoding can drive action without introducing decision paralysis. The phenomenological correlate is immediacy: the sense that the world simply is as it appears, without visible inferential distance between the signal and its evaluation. Immediacy is not a naive epistemic attitude; it is the phenomenological surface of action-forcing compression. Bounded systems under action pressure are organisationally forced into phenomenally closed modes of access — not because closure is chosen but because it is the only architecture compatible with coherent agency.

11.2 Selection Before Worldhood

Classical theories of perception assume a world that is first registered and then selectively processed. The F1 gating mechanism inverts this order. What the organism encounters is not a world from which it selects; it encounters only what has already passed the gate. The gate does not filter a pre-given world — it *constitutes* the operative world for this system at this moment. Selection is prior to worldhood, not posterior to it.

This is not a claim about the metaphysics of the external world. Physical signals exist prior to any gate. The claim is organisational: the world as available for evaluation, memory formation, action selection, and future expectation is the selection-conditioned subspace $S' = F_1(S)$. Everything outside S' is not a background that the system is aware of but ignoring; it is genuinely absent from the system's operative reality. The gate does not suppress signals; it determines what counts as a signal for this system in the first place.

A consequence of this inversion is that presence — the phenomenal sense of something being there — is better understood as stabilised exclusion than as faithful registration. When a signal is granted access through the gate, its presence in evaluation is constituted by the organised absence of everything that did not pass. The figure stands out against the ground not because the ground is perceived and suppressed but because the ground was never admitted. Presence is the positive face of a negative operation. To experience something as present is to inhabit a world in which an enormous amount has been organisationally excluded, and in which that exclusion is invisible — as it must be, by the argument of Section 10.2.

Certainty, in this framework, is a function of exclusion rigidity. A system with a high-confidence selection — one whose gate assigns very low probability to borderline signals — will experience its operative world as vivid, stable, and determinate. A system with a soft or uncertain gate will experience ambiguity, perceptual instability, or the sense that something is missing. The phenomenology of certainty is not epistemological; it is architectural. Certainty is high when the selection boundary is sharp and stable, regardless of whether the selected content accurately represents the distal environment. This provides a structural account of why certainty and accuracy can dissociate — confident misperception, anosognosia,

delusional conviction — without requiring separate explanatory mechanisms for each.

The formal chain connecting selection hardness to phenomenal certainty can now be stated precisely. High $H_s(G) \rightarrow$ strong exclusion $\rightarrow$ stable worldhood $\rightarrow$ phenomenal certainty. Each link in this chain is an organisational consequence, not a metaphysical postulate. High selection hardness means a large quantity of information is irreversibly destroyed at the gate. Strong exclusion means the operative world is sharply bounded — few signals are admitted, many are organisationally absent. Stable worldhood means the selection surface does not fluctuate: the same configuration of signals governs evaluation across cycles. Phenomenal certainty is the experiential signature of this stability: the world simply is as it appears, without visible alternatives crowding the margin of perception. The chain is formalizable, measurable, and — through the experimental programme of Paper 2B — partially empirically constrained. This is what it means to say that certainty is not an epistemic achievement but an exclusion geometry.

Presence, in this framework, is the deepest consequence of Selection Before Worldhood. A system that does not observe a world and then act in it, but whose world is constituted by the very selection that drives its action, cannot be a detached observer of its own experience. It is inside what it selects. World-model and action-loop are not two things coordinated — they are one closed dynamic, and presence is the name for that closure as encountered from within. Presence = enacted stabilized worldhood: not representation of a place but the organisational inseparability of what is selected and what is done with what is selected. A system without this inseparability — one that maintains a clean separation between its model and its action — is not an agent with presence; it is a planner with a map. The F1/F2 architecture, by making selection constitutive rather than preparatory, produces presence as an architectural consequence.

11.3 Actionable Compression Topology

Qualia have resisted functional analysis for a simple reason: every functional account attempts to express them in terms of relationships to other states, and every such account is met with the intuition that something is left out. The missing element is real, but it has been systematically misidentified. It is not a phenomenal residue floating above the functional order; it is the compression structure itself — the specific topology by which selected information is organised into an actionable format for downstream processing. We call this **Actionable Compression Topology**.

Consider what it means for a signal to be admitted through the F1 gate and delivered to the evaluation system. The signal does not arrive as raw data; it arrives as a structured encoding — urgency, polarity, temporal priority, interoceptive contribution — that has already organised the incoming information into a format directly usable by valence assignment and goal-alignment processing. This organisation is not arbitrary: it is a compression that preserves exactly what is needed for action and discards what is not. The compression is lossy in the information-theoretic sense, and irreversible: what is discarded at the gate cannot be recovered downstream.

The irreducibility of qualia, on this account, is not metaphysical but organisational. A quale — the redness of red, the painfulness of pain — is a compression singularity: a point in the system's representational topology at which multiple dimensions of signal information have been collapsed into a single actionable format, and from which the original multi-dimensional signal cannot be reconstructed. The quale is the compressed form, not the raw form. Attempts to explain it by unpacking it into its constituent dimensions

fail because the compression has already occurred; the analysis is performed on the output, not the input. This is why introspective accounts of qualia are always impoverished relative to the richness they report: the report is generated by the evaluation system operating on the compressed encoding, not by direct access to the compression process itself.

The explanatory gap is therefore the cost of the compression. Every lossy compression generates a gap between the original signal and its compressed representation — a gap that is not closable within the compressed domain. The hard problem is the philosophical discovery of this gap, mistakenly framed as a gap between the physical and the phenomenal rather than between the pre-compression signal and its actionable form. The correct framing: the explanatory gap is the information-theoretic cost of transforming a high-dimensional signal into an actionable singularity. It is not mysterious; it is expected. It is also ineliminable within any system that must act under resource constraints, because lossy compression is the price of actionability.

This reframing has a direct consequence for the debate over qualia and functional equivalence. Two systems with identical functional organisation — identical input-output profiles, identical valence responses, identical action selection — may nonetheless compress their signals through different topologies, producing different compression singularities. Whether those different singularities constitute different qualia, or whether functional equivalence implies phenomenal equivalence, is exactly the question the Artificial F1 test programme of Section 6.2 is designed to probe. The present framework does not answer it; it locates it correctly. The question is not whether the systems feel the same — it is whether their compression topologies are equivalent, and whether equivalence at the functional level entails equivalence at the compression level. That is an empirical question, not a philosophical one.

11.4 The Three-Level Architecture of Organizational Phenomenology

The three preceding sections — Phenomenal Inevitability, Selection Before Worldhood, and Actionable Compression Topology — converge on a single theoretical commitment: phenomenal character is not a free-floating addition to physical processing but a structural consequence of how bounded systems must be organised in order to act. This commitment defines a research programme we call **Organizational Phenomenology**: the systematic study of how phenomenal properties arise from, and are constrained by, the organisational architecture of information-selecting systems.

Organizational Phenomenology operates across three levels of analysis, each with its own explanatory targets and methods. At the **computational level**, the subject matter is the formal structure of selection: the topology of the gate, the dimensionality of the compression, the information-theoretic properties of the resulting encoding. Artificial F1 is the engineering body of this level — the formal architecture that implements selection, compression, and valence assignment in a computationally tractable form. The predictions of Theorem 1, Claim 2, and the test programme of Section 6.2 are the computational level's empirical programme. At the **cognitive level**, the subject matter is the consequences of organisational architecture for behaviour, learning, and experience: how selection-constituted worldhood shapes perception, memory, and action; how compression topology determines the structure of phenomenal reports; how opacity generates the sense of immediacy and givenness that characterises conscious experience. The F1/F2 framework, the E1/E2 distinction, and the misattribution hypothesis operate at this level. At the **phenomenological level**, the subject matter is the first-person structure of experience as

such: the felt quality of presence, the texture of certainty and uncertainty, the sense of agency and its limits. This level is not reducible to the cognitive or computational levels, but it is informed by them: the phenomenological structure of experience is shaped by the organisational architecture that produces it, even if it cannot be derived from that architecture by any known procedure.

The relationship between these three levels is the organisational analogue of the mind-body problem. Organizational Phenomenology does not claim to solve that relationship; it claims to locate it more precisely than existing frameworks. The hard problem, on this account, is the resistance of the phenomenological level to derivation from the computational level — a resistance that is predicted by the framework (opacity as stability shield) rather than treated as a brute fact. The gap between levels is not closed; it is explained. That distinction matters: an explained gap is a research programme; an unexplained gap is a permanent mystery.

The present paper establishes the computational level in formal detail and gestures toward the cognitive level through the F1/F2 architecture, the E1/E2 distinction, and the test programme. The phenomenological level is addressed primarily through the negative claim of Phenomenal Inevitability: what the phenomenological level must contain, given the organisational constraints of the system that generates it. A constructive account of the phenomenological level — specifying not merely what phenomenal character must be but how it arises from organisational architecture — is deferred to subsequent work. The parent thesis (Saka, 2026a) provides the phenomenological anchoring; the present paper provides the computational and architectural grounding. The two are designed to be read together.

Open research questions at each level are listed here as explicit targets rather than as gaps to be apologised for. At the computational level: Can compression topology be formally characterised in a way that predicts phenomenal similarity across systems? Is there a topology metric whose equivalence classes correspond to equivalence classes of phenomenal experience? At the cognitive level: Does the opacity boundary shift under learning, attention, or pharmacological modulation, and if so, how does the shift manifest in phenomenal reports? Is the E1/E2 transition a sharp threshold or a graded continuum, and what determines its location? At the phenomenological level: What is the minimal organisational complexity required for phenomenal character to arise? Does Phenomenal Inevitability apply to all information-selecting systems, or only to systems above some threshold of architectural complexity? These questions are not rhetorical. They are the research agenda of Organizational Phenomenology, and they are answerable — at least partially — by the empirical methods of cognitive science, computational modelling, and comparative architecture study.

The broader ambition of this framework is to provide a bridge between the engineering programme of Artificial F1 and the philosophical tradition that has treated phenomenal consciousness as permanently resistant to scientific treatment. That resistance is real, but it is not permanent. It is the resistance of a compression singularity to derivation from its pre-compression inputs — a structural property of the architecture, not a fundamental feature of reality. Organizational Phenomenology is the claim that this structure can be studied, characterised, and ultimately understood — not by dissolving the phenomenal into the physical, but by tracing the organisational path from physical constraint to phenomenal consequence. That path exists. The present work is an early map of it.

The three-level architecture maps onto a distinction that recent work in representational neuroscience has begun to articulate from a different direction. Pohl et al. (2024, 2026) argue that neural representation is not a unitary phenomenon but a multi-dimensional one, characterised independently along axes of sensitivity, specificity, independence, and sufficiency. Their framework provides tools for analysing representations within a given representational manifold. Organizational Phenomenology addresses what is prior to that manifold: the constitution of the representational space itself through selection-conditioned access. This is not a fifth dimension of their framework; it is the process that determines the space within which their four dimensions apply. At the computational level, constitution is the gating operator G that produces $S' = F_1(S)$. At the cognitive level, constitution is the stabilisation of the operative world: the system inhabits S' , not S . At the phenomenological level, constitution is presence: the felt givenness and immediacy of the operative world, which are the experiential signatures of the selection boundary the system cannot cross from the inside.

Terminological note. Readers of the companion technical paper (Saka, 2026b) will find the term representation constitution used in that context to name this same upstream process — the production of the selection-conditioned manifold S' prior to any encoding or evaluation. The present paper uses the term task constitution for the same operation viewed from above: the $F1$ gating operator does not optimise within a task; it constitutes the task-space by determining which signals count as relevant inputs at all. Task constitution (architectural level) and representation constitution (formal/mathematical level) are therefore two descriptions of the same upstream operation — the former emphasising its role relative to task specification, the latter emphasising its role relative to representational geometry. Pohl et al.'s framework operates downstream of both: their four dimensions (sensitivity, specificity, independence, sufficiency) characterise representations within a manifold that constitution has already produced. The present framework is not in competition with theirs; it addresses the level they presuppose.

This three-level account opens rather than closes the question of whether organisational structure is sufficient for phenomenal experience. If phenomenal organisation can emerge from selection, evaluation, and topological constraints — as the present framework proposes — then no additional metaphysical substance is required by default. The word 'default' is carefully chosen. The framework does not assert that substrate is irrelevant; it asserts that substrate-dependence must be demonstrated rather than assumed, by identifying which physical property contributes essentially to phenomenal character and why that property cannot be realised in a structurally equivalent but biologically different system. That question is now askable with the precision the framework provides: *if organisational equivalence is insufficient for phenomenal equivalence, which physical property contributes essentially, and why?* This is one of the central open questions of Organizational Phenomenology and, the present work suggests, of consciousness research as a whole.

11.5 Phase Transition and Phenomenological Closure

The formal architecture of Paper 2A provides an unexpected bridge between the engineering results and the phenomenological consequences developed in Sections 11.1–11.4. The bridge is the phase transition. In Paper 2A, the action-selection entropy $H(a)$ exhibits a discontinuity at the gating threshold τ^* : below threshold, the system produces high-entropy exploratory outputs; above it, entropy collapses and selection becomes committed. This is not merely a technical property of the architecture. It is, we argue here, the formal correlate of phenomenological closure.

Consider what happens at the transition point. Below τ^* , the system has not yet committed to a selection surface: multiple signal configurations are approximately equally weighted, and the operative world is

underdetermined. Above τ^* , the system has crossed into a selection-closed regime: a specific subset of signals constitutes the operative world, alternatives are organisationally excluded, and the gate has stabilised around a particular configuration. From inside the system, this transition has a phenomenological signature: the shift from ambiguity to certainty, from underdetermined perception to committed presence. The discontinuity in entropy is what uncertainty feels like when it resolves.

The formal connection can be stated precisely, though we flag it here as a theoretical extension rather than an established result. Let $H_s(G)$ be the selection hardness of architecture G , σ^2 the noise variance, and γ^* the collapse threshold from Proposition 2A-3.5. Then the condition $H_s(G)/\sigma^2 > \gamma^*$ — which determines that the system is in the collapse regime — is equivalent to the condition that the system is operating on a stabilised selection surface. Phenomenal inevitability (Section 11.1) predicts that this surface is opaque to the system: it cannot represent the full basis of its own selection. The phase transition is therefore not only an entropy event but an opacity event: the moment at which the system becomes phenomenologically closed. $\Delta H \rightarrow$ selection closure $\rightarrow$ certainty stabilization is the dynamical path from computational event to phenomenological consequence.

Three properties that phenomenologists have treated as primitive features of conscious experience — certainty, unity, and immediacy — emerge naturally from this framework as dynamical phase properties. Certainty is exclusion rigidity: the sharpness of the selection boundary above threshold. Unity is selection coherence: the fact that a single selection surface, not a set of competing surfaces, governs downstream processing after the transition. Immediacy is the absence of derivation: content admitted through a stabilised gate arrives in evaluation without a visible selection history, which is precisely the opacity predicted by Phenomenal Inevitability. None of these properties requires a separate phenomenological theory; all three are consequences of the phase transition architecture.

A second formal chain connects selection hardness to epistemic urgency in a way that has direct phenomenological consequences. As $H_s(G)$ increases, the information destroyed at the gate is irreversibly lost to downstream processing: alternative signal configurations become inaccessible, not merely de-prioritised. The formal consequence is that the system's epistemic flexibility — its capacity to revise its operative world by reconsidering excluded signals — decreases monotonically with $H_s(G)$. At the limit, a system with very high selection hardness has no revision capacity at the selection boundary: its operative world is as committed as the gate that produced it. The phenomenological correlate of this loss of flexibility is what we call **epistemic urgency**: the felt pressure toward commitment that arises when a bounded system under action pressure cannot afford to remain undecided. The chain $H_s(G) \uparrow \rightarrow$ alternative accessibility $\downarrow \rightarrow$ epistemic flexibility $\downarrow \rightarrow$ certainty pressure $\uparrow$ is not a metaphysical claim about the nature of certainty; it is a structural consequence of the gate geometry. Certainty, on this account, is not a cognitive achievement — it is an organisational inevitability for any system whose selection hardness exceeds the revision threshold. The hard problem of certainty — why committed belief feels settled rather than merely probable — has the same source as the hard problem of consciousness: opacity at the selection boundary.

The empirical grounding for this interpretation comes from EXP-1 in Paper 2B. The sharp performance discontinuity observed at the noise threshold σ^* — where classification accuracy collapses from above-chance to near-chance within a narrow σ interval — is precisely the behavioural signature of the phase transition predicted by Proposition 2A-3.5. Below σ^* , the system operates in the collapse

regime: $H_s(G)/\sigma^2 > \gamma^*$, selection is informationally committed, and performance is stable. Above σ^* , the regime inverts: noise overwhelms the selection signal, the gate loses its informativeness, and performance falls to chance. The kink in the EXP-1 curve is the empirical footprint of the entropy discontinuity. If the phenomenological reading of this section is correct, that kink is also the mathematical address of phenomenal ignition: the point at which a system of this architecture transitions from underdetermined access to committed presence. This is not offered as proof — the phenomenological interpretation is a theoretical extension, not a derivation from the data. It is offered as a research target: if certainty is a dynamical phase property, its onset should be measurable at precisely the σ^* identified by EXP-1, and manipulable by varying $H_s(G)$ and the safety margin m as specified in Section 4.3.3.

This interpretation is offered as a speculative theoretical extension. The formal results of Paper 2A establish the entropy discontinuity and the collapse regime; the phenomenological reading is an organisational interpretation of those results, not a derivation from them. What the interpretation does is locate a research programme: if certainty, unity, and immediacy are dynamical phase properties, they should be measurable — in the sense that the conditions under which they arise ($H_s(G)/\sigma^2 > \gamma^*$) are experimentally controllable, and the behavioural signatures of the transition (entropy discontinuity, selection stabilisation) are empirically detectable. Organizational Phenomenology is the claim that phenomenal properties of this kind are amenable to this form of investigation — not by reducing phenomenal character to physics, but by tracing the organisational path from architectural constraint to phenomenal consequence.

11.6 The Central Research Question

A terminological clarification is necessary before proceeding. This paper does not ask whether AI systems are conscious, nor does it attempt to resolve what consciousness is. Both questions remain open and are deliberately set aside. The question this paper does ask is more precise and more tractable:

Can selective, valence-grounded architectures reproduce the functional organisation associated with conscious experience — without resolving the phenomenal residue?

This reframing has a specific consequence: the hard problem does not block the engineering programme. F1 activation may or may not carry phenomenal character — that question is isolated at the E1 level and left open. What can be specified, implemented, and tested is whether an artificial system instantiating $F2 = \Phi(V, C, G, R, T, Au)$ over a structured valence signal field exhibits the functional signatures associated with selective, evaluative, temporally integrated cognition. Whether that constitutes 'real' consciousness in the phenomenal sense is a question this framework neither answers nor needs to answer in order to proceed. A fuller discussion of consciousness as an organizational continuum — including the relationship between conscious and unconscious mind, non-neural adaptive systems, and the limits of neurocentrism — is developed separately in *Beyond Binary Consciousness: Organizational Continuity Between Life and Mind* (Saka, 2026).

12. Conclusion

This paper has introduced Artificial F1 as a previously under-specified architectural layer in AI systems: a pre-evaluative, resource-constrained signal selection mechanism that operates prior to task definition and objective specification. The central claim is not a removal — that phenomenal consciousness is unnecessary — but an addition: current AI architectures lack an F1-equivalent layer, and supplying one provides a framework for selective, valence-grounded cognition that is designed to be more resource-efficient, more ecologically sustainable, and more structurally aligned with human cognitive organisation than systems that attempt to process all inputs uniformly.

The central empirical questions remain open. First: is the phenomenal character of F1 functionally separable from the information profile it delivers? If yes, artificial F1 is in principle implementable. If no, the hard problem reasserts itself as a practical engineering constraint. Second: can the signal dimensions and mechanistic substrate specified above — homeostatic urgency, survival-relevant polarity, temporal priority, interoceptive analog, habituation dynamics, gating, selective broadcast, and valence assignment — be implemented in a way that produces genuine functional equivalence with biological F1? The six-test programme of Section 6.2 — conflict resolution, urgency gradient, novelty response, misattribution probe, adversarial valence, and cross-architecture comparison — provides the falsification roadmap. The failure conditions of Section 6.3 identify where the model breaks. Together these constitute a research programme, not a solved problem.

The governance and alignment questions follow directly from the architecture. In valence-based systems, misalignment is not a failure of optimisation — it is a failure of moral encoding at the architectural level. This is a different problem from the one most alignment research addresses, and it requires a different intervention: not objective correction after deployment, but valence governance before it. The technical possibility of Artificial F1 does not determine how it will be used; that determination belongs to the institutional, political, and contractual structures within which it is developed and deployed.

The hard problem of consciousness remains unresolved. The present work suggests that its resolution may not be a prerequisite for building systems that reproduce the functional structure of cognition. If the engineering succeeds, it will provide indirect empirical support for the misattribution hypothesis of the parent thesis: that phenomenal character and functional information profile are separable, and that what we ordinarily call feeling is better understood as the evaluative output of the F2 system than as the raw presence of F1 itself.

This framework also provides a new lens on the Chinese Room. Rather than attempting to resolve Searle's original claim that syntax is insufficient for understanding, it reframes the debate in architectural and empirical terms. By distinguishing between phenomenal grounding (F1) and functional organisation (F2), the present account suggests that the central question is not whether a system *truly understands*, but whether it exhibits the structured, constraint-consistent behaviour typically taken as evidence of understanding. An Artificial F1 system is not a passive rule-follower over symbolic inputs: it actively selects, organises, and evaluates inputs under competing constraints — a structural transformation that the original Chinese Room scenario does not include. The framework further supports a misattribution hypothesis: observers may attribute understanding or consciousness based on stable F2-level behaviour, even in the absence of phenomenal grounding — suggesting that part of the intuition driving the Chinese Room argument may arise from conflating phenomenal presence with functional performance. In this sense, the Artificial F1 framework does not refute the Chinese Room argument; it transforms it into a

testable research programme. The burden therefore shifts from asserting the absence of understanding to specifying which functional criteria remain unmet — and whether those criteria depend on phenomenal experience at all. Alternative interpretations remain possible, particularly those that maintain a stronger link between phenomenal grounding and understanding (Searle, 1980); the present account does not claim to settle that debate but to provide a new empirical axis along which competing positions can be evaluated.

A machine that reasons well is a tool. A machine that evaluates, values, and adapts is something closer to an agent. The difference is not only technical. It is the difference between a hammer and a colleague — and the question of who the colleague works for, under what constraints, with what recourse for those affected by its decisions, and what becomes of the humans whose judgment it gradually displaces, is not a question that engineering alone can answer.

Intelligence is not determined solely by how much a system can process, but by how effectively it controls what it processes.

A note on method. The framework makes explicit what it explains, what it relocates, and what it leaves open. Every joint in the argument is visible. Readers who disagree know precisely where to push. The hard problem is not solved here — it is located: its source identified as the opacity that any bounded coherent system must maintain at its selection boundary. That location is itself a result. A framework that cannot be falsified is not a framework. This one has edges, and edges can be pressed.

Appendix A — Notation and Formal System Definition

A.1 Symbolic Conventions

Let $x_t \in S$ denote the system input (sensory or informational state) at discrete time t , where S is a measurable signal space. The following symbols are used throughout:

- $F_1: S \rightarrow \mathbb{R}^n$ — first-stage encoding function (Artificial F1)
- $F_2 \in \mathbb{R}^m$ — second-stage representational state
- $G(\cdot; \theta)$ — gating and transformation operator parameterised by $\theta \in \Theta$
- $\pi(\cdot)$ — policy / action selection operator
- $a_t \in A$ — action at time t ; A is the action space
- $\theta \in \Theta$ — parameter space of the gating function

A.2 Core Functional Architecture

The system is defined as a compositional mapping:

$$z_t = F_1(x_t) \quad F_2(t) = G(z_t; \theta) \quad a_t = \pi(F_2(t))$$

The full pipeline is therefore:

$$x_t \xrightarrow{F_1} z_t \xrightarrow{G} F_2(t) \xrightarrow{\pi} a_t$$

This compositional structure implements the Signal $\rightarrow$ Gating $\rightarrow$ Global Broadcast $\rightarrow$ Valence Assignment $\rightarrow$ Action Selection chain described in Section 5.6, with G subsuming both gating and broadcast stages, and π subsuming valence-weighted action selection.

A.3 Temporal Structure

Time is assumed discrete: $t \in \mathbb{N}$. State transitions may depend on history and stochastic perturbation:

$$x_t = \varphi(x_{t-1}, \varepsilon_t)$$

where ε_t is a stochastic perturbation term with distribution D_η . No assumption of Gaussianity is imposed.

A.4 Interpretation Constraint

The formal model does not assume explicit phenomenal representation, unified global workspace necessity, or introspective accessibility of F_2 . Only functional mappings are defined. Any mapping from formal objects to phenomenal states is interpretive, not axiomatic.

Appendix B — Model Assumptions and Scope Conditions

B.1 Structural Assumptions

Signal Decomposability. $x_t \in S$ is decomposable into representational features corresponding to the signal dimensions specified in Section 5.1–5.5 (urgency, polarity, temporal priority, interoceptive analog, habituation dynamics).

Functional Separation. F_1 and G are separable computational stages. No requirement of architectural co-localisation is imposed; the stages may be implemented in distributed substrates.

Finite Capacity. $\dim(F_1(x_t)) < \infty$ and $\dim(F_2(t)) < \infty$. Unbounded representation is not assumed.

B.2 Temporal Assumptions

The system is Markovian at the representation level:

$$F_2(t+1) \approx h(F_2(t), x_{t+1})$$

No long-term global memory is assumed unless explicitly modelled within the parameter space Θ .

B.3 Noise and Uncertainty

Observations are noisy: $x_t = \bar{x}_t + \eta_t$, where $\eta_t \sim D_\eta$ and noise is not assumed Gaussian. The gating threshold θ is adaptive under noise conditions as specified in Section 5.6.

B.4 Validity Domain

The model applies to: perceptual decision systems; cognitive architectures; agent-based policy systems; neural and hybrid computational systems.

The model does *not* claim validity for metaphysical accounts of consciousness or non-computational cognition models. Scope is strictly functional and architectural.

Appendix C — Extended Formalisation

C.1 Gate Function Decomposition

The gating operator G can be expressed as a parameterised nonlinear transformation. In the simplest case:

$$G(z_t; \theta) = \sigma(Wz_t + b)$$

where $W \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and σ is an activation function (sigmoid, ReLU, or task-appropriate variant). More generally, G may be any nonlinear function of z_t and θ satisfying the threshold and priority constraints of Section 5.6.

C.2 Evaluation Functional

Action selection is defined as utility maximisation over the F_2 state:

$$a^* = \operatorname{argmax}_{a \in A} U(a, F_2(t))$$

where $U: A \times \mathbb{R}^m \rightarrow \mathbb{R}$ is the utility functional. This formulation subsumes both the valence-weighted decision rule in Section 5.7 and the three assessment dimensions (value, salience, priority) of the parent thesis's Evaluation System.

C.3 Information Flow Constraint

Information flow from F_1 to F_2 is bounded by channel capacity:

$$I(F_1(x_t); F_2(t)) \leq C$$

where I denotes mutual information and C is the effective channel capacity of the gating and broadcast architecture. This constraint formalises the winner-take-all broadcast property of GNWT ignition described in Section 4.6: not all F_1 content reaches F_2 .

C.4 Reduction Principle

The system admits a reducibility condition expressing F_2 as a function of a temporal window of F_1 encodings:

$$F_2(t) = R(F_1(x_{t-k:t}), \theta)$$

This introduces temporal window dependence (k) and a compressed representation mapping R (a reduction operator). The equation is intentionally under-specified: its full dynamical, causal, and convergence-level expansion — including learning dynamics, robustness analysis, and adversarial perturbation conditions — constitutes the primary formal programme of subsequent work. The present appendix defines the architectural skeleton; formal analysis and empirical validation are deferred to that treatment.

This appendix establishes a minimal formal backbone and a bounded scope of interpretation for the $F_1 \rightarrow F_2$ architecture. It is not intended as a proof of correctness or completeness, but as a demonstration that the conceptual framework of this paper admits precise mathematical expression.

References

- Asimov, I. (1942). Runaround. *Astounding Science Fiction*. [First publication of the Three Laws of Robotics.]
- Birch, J. (2025). AI consciousness: A centrist manifesto. PhilArchive preprint. <https://philarchive.org/rec/BIRACA-4> [Epistemic caution on AI consciousness attribution: neither over-attribution nor dismissal. Cited in Section 10.2.]
- Butlin, P., & Lappas, V. (2025). There is no such thing as conscious artificial intelligence. *Humanities and Social Sciences Communications*. Nature Publishing Group.
- Cogitate Consortium (Ferrante, O. et al.). (2025). Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature*, 642(8066), 133-142. <https://doi.org/10.1038/s41586-025-08888-1> [Large-scale adversarial collaboration: IIT and GNWT predictions both partially disconfirmed. Cited as background evidence that existing dominant theories do not fully capture the organizational structure of conscious experience.]
- Damasio, A. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. Putnam.
- Dehaene, S. (2014). *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts*. Viking.
- Dehaene, S., Changeux, J.-P., & Naccache, L. (2011). The global neuronal workspace model of conscious access. In S. Dehaene & Y. Christen (Eds.), *Experimental and Theoretical Approaches to Conscious Processing* (pp. 55-84). Springer.
- European Commission. (2019). *Ethics Guidelines for Trustworthy AI*. High-Level Expert Group on Artificial Intelligence.
- European Commission. (2006). Regulation (EC) No 561/2006 on the harmonisation of certain social legislation relating to road transport. *Official Journal of the European Union*.
- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138.
- IEEE. (2019). *Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems*. First Edition. IEEE.
- Judge, M. (Director). (2006). *Idiocracy* [Film]. Ternion Pictures.
- Henriques, G., & Vervaeke, J. (2024). Splitting the hard problem of consciousness in two. *Psychology Today* [Blog post, Theory of Knowledge series]. Retrieved from <https://www.psychologytoday.com/us/blog/theory-of-knowledge/202411/splitting-the-hard-problem-of-consciousness-in-two>
- Henriques, G. (2024). The core of Extended Naturalism. *Medium / Unified Theory of Knowledge*. Retrieved from <https://medium.com/unified-theory-of-knowledge/the-core-of-extended-naturalism-fd1fc4fb887b>
- James, W. (1890). *The Principles of Psychology*. Henry Holt and Company. [Vol. 1, Chapter XI: Attention; the observation that experience tracks what is attended to.]
- Pohl, S., et al. (2024). Clarifying the conceptual dimensions of representation in neuroscience. *arXiv preprint*. [Four-dimensional framework: sensitivity, specificity, independence, sufficiency. Cited for the representation analysis / representation constitution distinction in Section 11.4.]
- Pohl, S., et al. (2026). Clarifying the conceptual dimensions of representation in neuroscience. *Nature Reviews Neuroscience*. [Peer-reviewed version of Pohl et al. (2024).]
- Kneer, M., & Viehoff, J. (2025). The hard problem of AI alignment: Value forks in moral judgment. *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*. ACM.

- Aru, J., Suzuki, M., & Larkum, M. E. (2020). Cellular mechanisms of conscious processing. *Trends in Cognitive Sciences*, 24(10), 814-825. <https://doi.org/10.1016/j.tics.2020.07.006> [Dendritic Integration Theory: layer 5 pyramidal neurons as coincidence detectors; apical/basal integration as the cellular basis of gating — cited as mesoscopic implementation candidate for F1 selection mechanism, Section 3.]
- Budson, A. E., Richman, K. A., & Kensinger, E. A. (2022). Consciousness as a memory system. *Cognitive and Behavioral Neurology*, 35(4), 263-297. <https://doi.org/10.1097/WNN.0000000000000319> [Memory Theory of Consciousness: empirical grounding for the temporal (T) and cognitive-symbolic (C) components of F2; dissociable deficits in episodic and working memory systems — cited as supporting evidence for F2 structure, Section 3. Note: MToC's priority ordering — memory preceding phenomenal presence — is not adopted; F1 precedes F2 in the present framework.]
- Hansen, T. (2025). Structural conditions for momentary consciousness. SSRN preprint. <https://ssrn.com/abstract=5352852> [Structural ignition model: four jointly sufficient conditions for conscious ignition (internal state awareness, temporal binding, symbolic referencing, causal output from self-model); consciousness as discrete event not persistent state — cited for structural mapping to F1/F2 architecture, Section 3 and companion document *hansen_f1f2_mapping*. Note: preprint, not peer-reviewed at time of citation.]
- Nagel, T. (1974). What is it like to be a bat? *Philosophical Review*, 83(4), 435-450.
- Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417-424.
- OECD. (2019). *OECD Principles on AI*. OECD/LEGAL/0449.
- Saka, H. (2026a). *Toward a Reframing of the Hard Problem of Consciousness: Subjective Reality, Feeling, and the Origins of the Conceptual World*. Version 7.18. [Independent academic monograph].
- Saka, H. (2026b). *Artificial F1: Full Computational Model — Selection Hardness, Non-Scalarizability, and Phase Transition in Bounded Evaluative Architectures*. Zenodo. <https://doi.org/10.5281/zenodo.20524004>
- Saka, H. (2026c). *Hard-Gating Collapse Dynamics: Selection Hardness as the Organizing Parameter for Robust Sparse Routing*. Zenodo. <https://doi.org/10.5281/zenodo.20523503>
- UNESCO. (2021). *Recommendation on the Ethics of Artificial Intelligence*. UNESCO General Conference, 41st session.

This paper is a companion to: Saka, H. (2026). *Toward a Reframing of the Hard Problem of Consciousness: Subjective Reality, Feeling, and the Origins of the Conceptual World*. Version 7.18. All theoretical claims, structural decisions, and intellectual commitments are the author's own. The paper was developed with the assistance of AI language models as interlocutors for argumentation and critical review.